

Brain Before the Swipe: Can Consumer-Grade EEG Predict Viewer Engagement during Naturalistic TikTok Browsing?

Master's Thesis

Gregor Lederer

Supervisors:

Dr. Irina-Catrinel Craciun

Prof. Dr. Dr. h. c. mult. Lothar H. Wieler

Prof. Dr. Bert Arnrich

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Hasso Plattner Institute
University of Potsdam

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Abstract

Short-form video platforms such as TikTok structure user attention as a rapid sequence of binary micro-decisions: continue watching, or swipe away. Platform metrics and much of the surrounding public discourse treat continuous, uninterrupted viewing time as a proxy for user engagement, yet recent behavioral evidence suggests that prolonged, uninterrupted scrolling is at least as consistent with passive, dissociative consumption as with effortful, goal-directed attention. This thesis investigates whether electroencephalography (EEG), recorded unobtrusively during naturalistic short-form video browsing, can discriminate the neural state preceding a voluntary skip decision from the neural state during sustained, non-skip viewing, and whether any such discrimination exceeds what is achieved by the conventional Engagement Index ($\beta/(\alpha + \theta)$), a composite band-power ratio drawn from clinical and educational neuroscience.

Twenty-five participants ($n = 25$; 13 female, 12 male; age $M = 24.6$, $SD = 2.9$) were recorded with a consumer-grade, four-channel Muse S headband (AF7, AF8, TP9, TP10) while freely browsing a personal TikTok “For You” feed for 25 minutes in a private, distraction-free setting, following a two-minute resting baseline. Manual swipe events were synchronized to the EEG stream via keypress markers. A literature-motivated labeling scheme operationalized two classes: **SKIP**, drawn from a 3.0-second imminent-skip period ending 2.0 seconds before each swipe (a gap exceeding both the reported onset of the Bereitschaftspotential and directly measured pre-movement EMG onset in comparable self-paced tasks), and **STAY**, comprising all remaining continuous, non-boundary viewing. From this labeling, 236 univariate features spanning seven frequency bands, four electrodes, and the raw broadband signal were engineered, including band power statistics, Hjorth parameters, and two novel time-domain frequency descriptors (Peak Frequency, Macro Frequency).

Class separability, quantified via Cohen’s d , was consistently higher for individual frequency bands, electrodes, and statistical descriptors than for the Engagement Index; the single strongest feature, raw-signal standard de-

viation at TP10, exceeded the Engagement Index’s separability by nearly tenfold. In logistic regression benchmarking against a coin-flip baseline, the Engagement Index achieved above-chance but class-asymmetric performance, detecting **STAY** states substantially better than **SKIP** states. Three candidate features identified through the Cohen’s d ranking — TP10 raw standard deviation, AF7 high-gamma mean, and AF8 delta peak frequency — each yielded statistically significant improvements over the Engagement Index in **SKIP** detection at both the intra-subject and inter-subject level, though absolute performance remained modest and, in the cross-participant setting, did not exceed the chance-equivalent reference. A secondary behavioral analysis further showed that swipes cluster into rapid, individually variable “burst” sequences rather than occurring as uniformly spaced discrete events.

These findings indicate that the conventional Engagement Index captures only a limited, asymmetric component of the neural signal associated with short-form video browsing, and that more granular, region- and frequency-specific EEG features carry additional, currently underexploited discriminative information about imminent disengagement. The thesis further contributes a fully reproducible, open-source recording and analysis pipeline for consumer-grade EEG in naturalistic digital-media settings, and discusses the neuroscientific, psychological, behavioral, and public-health implications of these results, while explicitly flagging electromyographic and line-noise contamination as unresolved confounds requiring artifact-controlled replication.

Chapter 1

Introduction

1.1 Motivation

Short-form video platforms have become one of the dominant modes of digital media consumption worldwide, and TikTok in particular has popularized an interaction paradigm in which content is delivered as an effectively infinite, algorithmically curated stream, navigated by a single recurring behavioral unit: the swipe. Each swipe is the outward expression of a micro-decision — to continue watching, or to move on — repeated hundreds of times within a single session. Understanding what happens, neurally and behaviorally, in the seconds preceding this decision is of interest not only to platform design and human–computer interaction research, but increasingly to public health communication, which must compete for exactly this attentional currency when delivering health messages, educational content, or behavior-change interventions through the same channels.

The default proxy for engagement used both by platforms and by much of the surrounding research and policy discourse is duration: how long a user continues watching before swiping away. This proxy is intuitive and trivially measurable from platform telemetry, but its behavioral validity is contested. Attention research using cued-recall paradigms has shown that sustained attention is better characterized as a fluctuating, trial-by-trial internal state, decodable from prestimulus EEG activity, than as a fixed property that scales linearly with time-on-task [1]. More directly relevant to the present context, two recent studies interrogating social media use specifically have found that extended, uninterrupted engagement does not reliably track productive attentional states: spontaneous flow experiences are rarely attributed to social media use even during heavy use [2], and extended, uninterrupted scrolling sessions are strongly associated with normative dissociation — users becom-

ing “lost” in the feed, unable to later recall what they consumed [3]. Taken together, this literature suggests that continuous non-skip viewing time is an ambiguous behavioral signal, equally consistent with effortful engagement and with passive, low-encoding consumption.

This ambiguity motivates a different empirical question, one this thesis takes as its central object of study: rather than assuming that duration indexes engagement, can the moment of disengagement itself — the decision to skip — be characterized directly at the level of the EEG signal, independent of any prior assumption about what continuous viewing means? If a decodable neural signature precedes the voluntary skip action, it would offer a behaviorally grounded, physiologically anchored complement to duration-based metrics, and a candidate target for evaluating whether existing composite engagement measures, such as the Engagement Index ($\beta/(\alpha + \theta)$) widely used in educational and human-factors neuroscience, actually capture the process relevant to short-form video interaction.

1.2 Research Questions

This thesis addresses three interrelated research questions:

1. **Separability:** Can EEG recorded during naturalistic, self-paced TikTok browsing be used to distinguish the period preceding a voluntary skip decision (SKIP) from continuous, non-skip viewing (STAY), using only a lightweight, consumer-grade, four-channel headband?
2. **Benchmark comparison:** How does the class-separation and predictive performance of individual, granular EEG features compare against the conventional Engagement Index, both as a descriptive effect-size measure (Cohen’s d) and as a predictive model (logistic regression), at both the intra-subject and inter-subject level?
3. **Parameter sensitivity:** How sensitive is any observed separability to the specific temporal parameters — window length, pre-swipe safety gap, and imminent-skip period — used to define the STAY/SKIP labeling scheme itself?

Answering these questions requires solving a chain of non-trivial methodological problems that do not have an established, off-the-shelf solution in the literature: a naturalistic recording protocol that preserves ecological validity while achieving millisecond-level behavioral synchronization; a principled, literature-motivated labeling scheme for a self-paced (rather than externally

cued) behavioral event; a feature space broad enough to allow genuinely comparative evaluation against a composite clinical baseline; and a cross-validation and significance-testing framework robust to the strong temporal autocorrelation and severe class imbalance inherent to naturalistic swipe data. The bulk of this thesis is accordingly devoted not only to answering the three questions above, but to building and documenting a reusable methodological pipeline capable of doing so reproducibly.

1.3 Approach

To address these questions, twenty-eight participants were recruited, of whom twenty-five ($n = 25$) were retained for analysis after exclusion of compromised or ineligible recordings (Section 3.1). Each participant was recorded with a consumer-grade Muse S EEG headband while freely browsing their personal TikTok feed for 25 minutes in a private, isolated setting designed to elicit naturalistic viewing behavior, preceded by a two-minute resting baseline used for per-participant signal normalization (Sections 3.2–3.3). Manual swipe events were synchronized to the continuous EEG stream via a custom keypress-logging architecture built on the Lab Streaming Layer protocol (Section 3.5).

A three-parameter labeling scheme was then developed to derive **STAY** and **SKIP** classes from this continuous, self-paced behavioral stream, with default parameter values grounded in the EEG window-length literature and in the Bereitschaftspotential and pre-movement EMG literature governing motor-preparatory contamination near the swipe action itself (Sections 3.4–3.4.2). Because none of these parameters had established precedent for a self-paced, naturalistic swiping paradigm specifically, their influence on downstream class separability was additionally characterized through a one-at-a-time sensitivity analysis (Section 3.4.4) rather than assumed to be optimal by default.

From the resulting labeled dataset, a broad feature space of 236 univariate EEG descriptors was engineered per 1.0-second analysis window, spanning conventional band-power statistics, Hjorth parameters, and two novel time-domain frequency descriptors, alongside features drawn from the undecomposed raw broadband signal (Section 3.7). Feature-level and category-level class separability was quantified via Cohen’s d and benchmarked directly against the Engagement Index (Section 3.4.5), and the strongest candidate features were subsequently evaluated as standalone logistic regression classifiers against both a chance baseline and the Engagement Index baseline, under both intra-subject (blocked temporal cross-validation) and

inter-subject (leave-one-participant-out) evaluation, with Wilcoxon signed-rank testing used to assess statistical significance across the participant cohort (Sections 3.8.3–3.8.4).

1.4 Contributions

This thesis makes five principal contributions, each developed in detail in Chapter ??:

1. A fully naturalistic, ecologically valid EEG experiment synchronizing consumer-grade EEG with free TikTok browsing behavior in $n = 25$ participants, together with a reproducible experimental protocol (Section 4.1).
2. An open-source, reusable recording and analysis pipeline covering automated EEG acquisition, behavioral synchronization, artifact-aware pre-processing, feature engineering, and statistically rigorous model evaluation, transferable to related naturalistic EEG paradigms beyond the present study (Section 4.2).
3. A characterization of swipe-burst behavior, showing that skip events cluster into rapid, individually heterogeneous sequences rather than occurring as uniformly spaced discrete decisions (Section 4.3).
4. A direct, statistically grounded comparison between the conventional Engagement Index and granular, feature-level EEG descriptors in their ability to separate and predict **STAY** versus **SKIP** states, identifying specific features (raw TP10 variability, AF7 high-gamma activity, AF8 delta peak frequency) that outperform the Engagement Index despite the latter’s established status as a clinical and educational benchmark (Sections 4.4–4.4.2).
5. A neuroscientific, psychological, and behavioral interpretation of these results, together with an explicit, appropriately scoped discussion of their implications for public health communication, digital media policy, and health message design, and of the artifact- and hardware-related confounds that limit the strength of any causal claim (Section 4.5).

1.5 Thesis Structure

The remainder of this thesis is organized as follows. Chapter 2 (Literature and State of the Art) situates the present work within the literature on attention and engagement in short-form video, sampling-boundary handling and parameter sensitivity in EEG classification, class-separation metrics, time-domain frequency estimation, and EMG and line-noise contamination in consumer EEG. Chapter 3 (Materials and Methods) details the participant sample, data acquisition hardware and protocol, labeling strategy and its literature grounding, preprocessing pipeline, feature engineering, and the statistical framework used for model evaluation. Chapter 4 (Results and Discussion) reports the five contributions summarized above and closes with a discussion of their neuroscientific, psychological, behavioral, and applied implications, together with explicitly flagged limitations. The thesis concludes with a set of concrete directions for future research, spanning validation and replication, mechanistic artifact clarification, theoretical development of the engagement construct, methodological expansion, and application to digital public health.

Chapter 2

Literature and State of the Art

The following sections outline the core theoretical claims, methodologies, and prior research required to contextualize this study.

2.1 Adjacent studies

- **"Detection of Short-Form Video Addiction with Wearable Sensors Via Temporally-Coherent Domain Adaptation"** [4] - Same with us: Muse S, TikTok, Freq.Band wise performance, Muse S sampling rate of 256 Hz - Different to us: Addiction (difference to us) instead of engagement, LSTM instead of Cohen d and LogReg

2.2 Libraries

- the `muselsl` library [5]

2.3 Engagement and Attention in short-form video platforms

The behavioral status of continuous, uninterrupted viewing on short-form video platforms is contested in the literature. Attention research using cued recall paradigms has shown that sustained attention is a fluctuating internal state — trial-by-trial, moment-to-moment variation in an EEG-decodable prestimulus signal that predicts subsequent encoding — rather than a fixed property that scales with time-on-task [1]. This distinction matters because two recent HCI studies suggest that longer, uninterrupted engagement with

social media specifically does not track this kind of productive sustained attention, and may in fact index its absence. In a five-day field study pairing app telemetry with unprompted daily reconstructions, Knierim et al. found that participants spontaneously attributed flow experiences to social media in only 2% of reported instances, and that within-person increases in social media use — particularly passive, feed-based use — predicted fewer subsequent flow reports and lower momentary readiness [2]. Complementarily, Baughan et al. deployed a custom Twitter client and found that extended, uninterrupted scrolling sessions were strongly associated with normative dissociation: users reported becoming "lost" in the feed, losing track of time, and later being unable to recall what they had consumed, with interface features that segmented or interrupted continuous viewing (custom lists, reading-history labels, time-limit prompts) significantly reducing self-reported dissociation [3]. Taken together, this body of work indicates that duration of continuous, non-skip viewing is an ambiguous behavioral signal: it is equally consistent with effortful, goal-directed engagement and with passive, dissociative consumption in which little of the viewed content is encoded. Our **STAY** class therefore should not be interpreted as a direct behavioral proxy for sustained cognitive engagement; rather, it captures continuous non-skip viewing time, a necessary but not sufficient behavioral correlate of engagement that the prior literature suggests may just as often reflect disengaged, dissociative viewing.

A further methodological question concerns whether duration-based behavioral proxies for engagement can be validated, or at least cross-checked, against physiological markers, and here the evidence is more cautionary than supportive. Recent work directly contrasting EEG spectral markers of sustained attention against markers of cognitive engagement in a naturalistic learning context found that established engagement-sensitive markers — notably frontal midline theta — failed to discriminate between a sustained-attention task and a cognitive-engagement task, whereas posterior beta, occipital alpha, and the classical $\beta/(\alpha + \theta)$ ratio did discriminate reliably [6], though the authors themselves attribute this null result partly to a baseline confound (a post-effort baseline potentially saturated with residual fatigue) and a modest sample ($n=15$), rather than presenting it as an unqualified failure of frontal midline theta as an engagement marker. The same study further showed that these markers followed distinct, non-linear (U-shaped and inverted-U-shaped) trajectories over the course of a continuous naturalistic task, such that aggregating spectral power over an entire viewing period — the same logic underlying a duration-based **STAY** label — obscures exactly the real-time fluctuations that separate momentary attentional states from one another. Taken together with the behavioral literature reviewed

above [1–3], this indicates that no existing marker, behavioral or neural, has been shown to positively validate continuous non-skip viewing time as a proxy for sustained cognitive engagement; the available evidence instead consistently undermines duration as a proxy, whether via dissociative viewing states, the rarity of flow during passive consumption, the trial-by-trial (rather than duration-locked) nature of attentional fluctuation, or the failure of aggregate engagement markers to track task state. The present thesis therefore does not treat the **STAY** label as validated evidence of engagement; it instead adopts it as an explicit, literature-motivated behavioral operationalization, and treats the subsequent EEG analysis (Methods Sections 3.7–3.8) as an independent empirical test of what, if anything, separates **STAY** from **SKIP** at the level of the signal – rather than as a confirmation of the label’s validity.

2.4 Sampling Boundaries in EEG and physiological-signal classification

Boundary handling during epoch labeling is a recurring but often understated methodological choice in EEG- and physiological-signal classification. Rather than allowing a fixed window to span two adjacent labeled events, Moontaha et al. [7] process incoming EEG in non-overlapping tumbling windows and assign a class label to a window only when it falls entirely within a single stimulus interval; windows that straddle the boundary between two labeled segments are excluded from training rather than assigned to either class. This design choice reflects a broader assumption in the field: because short-timescale physiological signals are treated as stationary only within narrow intervals, and because the true onset of a cognitive or behavioral event relative to its recorded label is inherently uncertain, epochs near event boundaries are a primary source of label noise. Excluding them is preferred over resolving the ambiguity through interpolation or majority-vote heuristics, since misassigned boundary epochs directly degrade class separability during training. For the present work, this precedent motivates treating boundary overlap between **STAY** and **SKIP** not as an edge case to be smoothed over, but as a structural feature of swipe-event labeling that should be excluded outright, even at the cost of reduced sample count.

2.5 Parameter Choice in EEG-classification

Existing evidence on temporal parameter choice in EEG-based classification is concentrated on two variables – window length and label latency –

and is generally obtained by varying one parameter at a time rather than jointly. Window lengths between 1 and 5 seconds have been tested on both a benchmark dataset (AMIGOS) and wearable-EEG recordings, with a general decline in F1-score as window length increased across affect dimensions, classifiers, and devices, and 1-second windows performing best [7]. The effect of label latency has been isolated by artificially delaying the availability of ground-truth labels by the mean stimulus duration (86 s) while holding the rest of the pipeline fixed: this alone reduced mean arousal F1-score by roughly 20 percentage points (from above 0.82 to 0.637), and left valence classification below the level reported as chance [7]. In motor-imagery decoding, time window and frequency bandwidth had previously been optimized individually but never jointly; addressing this with a full grid search across five time windows and 23 frequency bands showed both parameters to significantly affect classification accuracy (repeated-measures ANOVA, $p < 0.001$ for each), while their interaction was nominally significant before correction ($p = 0.044$) but did not survive the study’s Bonferroni threshold, and the population-level optimal combination did not match the individually optimal combination for many participants [8]. Read together, these studies indicate that temporal parameter choice can substantially affect EEG-classification performance on its own, that label latency in particular can be sufficient to collapse performance toward chance, and that one-at-a-time variation is the prevailing approach for characterizing these effects, notwithstanding its recognized inefficiency relative to designs drawn from statistical experimental theory [9]. None of these studies, however, examine parameters governing the temporal placement of a label relative to a naturalistic, self-paced behavioral event rather than an externally imposed experimental cue – the question addressed by the sensitivity analysis reported below.

Beyond the window-length and label-latency sweeps already discussed, two further strands of evidence bear on the direction of the effects examined in this thesis. First, on window length: Tzimourta et al. segmented EEG epochs into eight discrete lengths from 5 s to 12 s for the detection of Alzheimer’s Disease and found that classification accuracy increased step-wise with window length across all six classification problems and five brain regions tested, with no reversals [10]. This provides an independent, monotonic precedent for the intuition that longer windows – carrying more samples per estimate – yield more separable features, in a paradigm unrelated to short-form video but sharing the underlying signal-processing logic. Second, on the placement of a measurement window relative to an upcoming event: Rimbert et al. varied the temporal position of the baseline window relative to the next motor-imagery trial and found that baselines positioned closer to the end of the preceding trial (i.e. further from the upcoming cue) pro-

duced significantly stronger, more spatially global event-related desynchronization and higher classification accuracy than baselines positioned closer to the cue [11]. This is structurally the same question as the gap-length and imminent-skip-period parameters examined here: both ask whether moving a labelled window further from a to-be-predicted motor event improves the separability of that label. Taken together, these two studies motivate the expectation that, in the present paradigm, larger windows and greater temporal distance from the swipe event should both improve class separability. However, this expectation should not be treated as a universal law of EEG parameter selection: Mi et al., sweeping window length for motor-imagery classification across nine subjects, found that the accuracy-optimal window size differed from subject to subject and was not monotonic within a single subject, with differences of up to 13% between window choices [12]. This indicates that window-length and event-distance effects, while directionally supported in the aggregate, can still be idiosyncratic at the individual level – precisely the caveat carried forward into the parameter sensitivity analysis in Section 3.4.4, where gap length, window length, and imminent-skip period are swept one at a time and the resulting trends are reported descriptively rather than optimized into a final parameter choice.

2.6 Sampling Boundaries in EEG and Physiological-Signal Classification

This same boundary-handling logic directly informs the present pipeline’s windowing procedure. Moontaha et al. process incoming EEG in non-overlapping tumbling windows and assign a class label to a window only when it falls entirely within a single stimulus interval, excluding windows that straddle the boundary between two labeled segments rather than resolving the ambiguity through interpolation or majority-vote heuristics [7]. The present pipeline adopts the same principle at the point of window extraction: 1.0-second analysis windows are drawn with no overlap (stride equal to window length) specifically so that no window can span a class boundary, a safety gap, or an adjacent sample, mirroring the tumbling-window exclusion strategy used for stimulus-interval labels. Because short-timescale physiological signals are treated as stationary only within narrow intervals, and because the true onset of a behavioral event relative to its recorded label is inherently uncertain, this precedent motivates excluding boundary-adjacent epochs outright in the present swipe-event paradigm as well, even though the source study used externally-cued stimulus intervals rather than self-paced

swipe events.

Boundary handling during epoch labeling is a recurring but often understated methodological choice in EEG- and physiological-signal classification. Rather than allowing a fixed window to span two adjacent labeled events, Moontaha et al. [7] process incoming EEG in non-overlapping tumbling windows and assign a class label to a window only when it falls entirely within a single stimulus interval; windows that straddle the boundary between two labeled segments are excluded from training rather than assigned to either class. This design choice reflects a broader assumption in the field: because short-timescale physiological signals are treated as stationary only within narrow intervals, and because the true onset of a cognitive or behavioral event relative to its recorded label is inherently uncertain, epochs near event boundaries are a primary source of label noise. This is consistent with the label noise literature more broadly: Frénay and Verleysen [13] show that mislabelling is not uniformly distributed across a dataset but is systematically more probable for samples near the classification boundary or in regions where classes are difficult to separate — a noisy-not-at-random (NNAR) pattern rather than a noisy-completely-at-random one. Under this model, label noise concentrated at class boundaries is disproportionately harmful: it not only degrades classification accuracy but increases the number of training samples required to reach a given level of reliability, meaning that a smaller but purer set of boundary-excluded samples can outperform a larger set containing boundary-adjacent noise. Excluding boundary epochs is therefore preferred over resolving the ambiguity through interpolation or majority-vote heuristics, since misassigned boundary epochs directly degrade class separability during training, and the resulting loss in sample count is expected to be offset by the reduction in exactly the type of noise most costly to classifier performance. For the present work, this precedent motivates treating boundary overlap between **STAY** and **SKIP** not as an edge case to be smoothed over, but as a structural feature of swipe-event labeling that should be excluded outright, even at the cost of reduced sample count.

2.7 Sensitivity to Parameter Choice in EEG-based classification pipelines

Methodologically, sensitivity to parameter choice in EEG-based classification pipelines is most commonly assessed through one-factor-at-a-time (OAT) variation, in which a single parameter is varied while all others are held fixed [14]. Despite its widespread use, this approach is known to be statisti-

cally inefficient and cannot identify interaction effects between parameters, since any two OAT sweeps only ever probe the space along axis-aligned lines through a single fixed operating point [9]. Nonetheless, OAT sweeps of individual temporal parameters have established precedent in EEG decoding: window length has been varied in isolation across a 1–5 s range while holding the remainder of the pipeline fixed, showing a general decline in classification performance as window length increases [7], and label latency has been varied in isolation by artificially delaying label availability, which alone reduced performance by roughly 20 percentage points [7]. Where the joint effect of two parameters has instead been assessed via full factorial grid search, both parameters were found to significantly affect classification accuracy on their own, but their interaction did not survive correction for multiple comparisons [8], illustrating both the value and the limited scope of interaction analysis relative to OAT. No existing study, to our knowledge, applies OAT to parameters governing the temporal placement of a label relative to a self-paced behavioral event, as opposed to an externally cued or delay-based one; the gap length and imminent-skip period examined below extend the same logic of temporal-parameter sensitivity to this self-paced setting.

2.8 Default Labeling Parameter Selection

Choosing the temporal parameters that define the **STAY**/**SKIP** labeling scheme – window length, gap length, and imminent-skip period – requires grounding in existing evidence on EEG window sizing and on the timing of pre-movement neural activity, since these two literatures bear directly on two of the three parameters while the third has no comparable precedent. On window length, Moontaha et al. swept window sizes from 1 to 5 s on wearable-EEG affect classification and found a general, monotonic decline in F1-score as window length increased, with 1-second windows performing best across affect dimensions, classifiers, and devices [7]. This motivates the present thesis’s choice of a 1.0-second default window, rather than a longer window that would risk diluting the rapid, self-paced cognitive transitions this paradigm targets. On gap length, the pre-swipe exclusion window must be wide enough to exclude motor-preparatory activity from the **SKIP** class: the *Bereitschaftspotential*, the best-characterized EEG correlate of self-paced voluntary movement, begins as an early, slow-rising negativity approximately 1.5 s before movement onset, with a steeper late component emerging 400–500 ms before onset [15, 16]. A 2.0-second gap therefore exceeds the full reported extent of the *Bereitschaftspotential* with a safety margin, ensuring that windows labeled **SKIP** reflect pre-decision cognitive state rather than the

motor execution of the swipe itself. No comparable literature exists, however, for the imminent-skip period specifically in the context of a self-paced, naturalistic swiping paradigm; unlike window length and gap length, this parameter cannot be anchored to an established finding and is therefore treated in this thesis as a literature-motivated but ultimately heuristic starting value, addressed directly through the one-at-a-time parameter sensitivity analysis in Section 3.4.4 rather than through a fixed literature-derived justification.

2.9 Class Separation Metric

Quantifying how well a candidate label definition or feature separates two behavioral classes requires an effect-size measure that is standardized and scale-invariant, since raw mean differences are not comparable across features expressed in heterogeneous units (band power, frequency counts, raw voltage) or across labeling parameterizations that alter sample size. Cohen’s d , the standardized mean difference between two distributions expressed in pooled-standard-deviation units, was developed precisely to meet this requirement: by dividing the raw effect by the common within-population standard deviation, it yields a "pure," dimensionless index that is comparable regardless of the original measurement scale [17]. This property is what makes it suitable here, where the same separability question – how distinguishable are two classes – is asked repeatedly across quantities as different as raw broadband voltage, band-relative power, and peak-frequency counts. Cohen’s d has established precedent specifically in EEG-based group- and class-separability analysis: Henao Isaza et al. use it to rank and select EEG features by the magnitude of the difference between an at-risk and a healthy control group prior to model training [18], and Kiarashi et al. use it to quantify how strongly a classifier’s predicted probabilities separate a clinical from a non-clinical group, explicitly reporting its improvement as a marker of increased class discriminability [19]. Both applications mirror the two uses made of the metric in the present work: ranking individual features by their standardized separation between **STAY** and **SKIP** (Section 3.8.2), and tracking how that separation shifts under different labeling parameterizations (Section 3.4.4). This precedent motivates treating Cohen’s d , rather than a raw or model-performance-based criterion, as the primary descriptive currency for separability throughout the labeling-strategy and feature-ranking pipeline.

Cohen’s d is the standard scale-invariant index for quantifying the standardized difference between two distributions, expressing a mean difference in units of pooled standard deviation rather than in the arbitrary units of the underlying measure [17]. This property is what makes it suitable for

comparing separability across heterogeneous feature types — spectral power, entropy, connectivity, or raw amplitude — that would otherwise be incommensurable on their native scales. The metric has established precedent specifically in EEG-based group and class separability analysis: Henao Isaza et al. use Cohen’s d to rank EEG-derived spectral, connectivity, and entropy features by their ability to separate Alzheimer’s-risk carriers from healthy controls, treating it as a principled feature-selection criterion prior to classifier training [18]. Kiarashi et al. apply the same metric to quantify group separability between MCI and cognitively normal participants directly from model output distributions, and further show that separability under Cohen’s d responds systematically to methodological choices — in their case, multi-night signal aggregation — with (d) increasing alongside statistical significance as aggregation improved [19]. Taken together, this precedent establishes two points relevant to the present work: first, that Cohen’s d is an appropriate and validated tool for ranking discriminative EEG features independent of their native units, motivating its use here for feature-level class-separation ranking (Section 4.4.1); and second, that Cohen’s d is equally appropriate as a sensitivity metric for tracking how a labeling or preprocessing parameter choice affects downstream separability, motivating its use here for the labeling-parameter sensitivity analysis (Section 3.4.4). Neither prior use, however, involves a self-paced behavioral labeling scheme comparable to the STAY/SKIP paradigm applied here — extending this precedent to that specific setting is a contribution of the present work.

2.10 Peak Frequency as a Time-Domain Frequency Estimator

Estimating a signal’s frequency content directly in the time domain, without an intervening Fourier transform, has established precedent along two lines. Hjorth’s Mobility parameter, one of the earliest and most widely adopted time-domain EEG descriptors, is defined as the square root of the ratio between the variance of a signal’s first derivative and the variance of the signal itself, and is explicitly derived to correspond to the standard deviation of the power spectrum — i.e., a formally grounded mean-frequency estimate obtained without computing that spectrum [20]. A second, independent line of precedent is the zero-crossing rate, which counts sign changes per unit time and has established use as a time-domain frequency proxy in speech and EEG classification pipelines [21]. Both methods are validated in the sense that their relationship to the underlying frequency content is derived

analytically (Mobility, via spectral moments) or holds exactly for simple periodic signals (zero-crossing rate, for a pure sinusoid). Peak Frequency, as used in the present thesis, follows the same general premise – that structural features of the raw time-domain signal encode frequency information – but operationalizes it as a direct count of local maxima per unit time rather than a derivative-variance ratio or a sign-change rate. This specific operationalization has not, to our knowledge, been independently validated against ground-truth frequency in prior work; the closest analogous claim in the literature, that local extrema are informative of a signal’s "global frequency" and function as a down-sampled representation of it, is asserted without independent validation in nonlinear time-series analysis [22]. Peak Frequency is therefore treated in this thesis as a heuristic, self-defined descriptor motivated by this broader premise rather than as an established measure, and its discriminative value is instead established empirically via the feature-ranking ablation in Section 4.4.1.

2.11 EMG Artifacts in EEG

Temporal electrode sites are not neutral with respect to artifact risk: they sit directly over the temporalis muscle, one of the two dominant sources of myogenic contamination in scalp EEG, and this contamination is broadband, present even at rest, and not reliably removed by standard correction methods. Goncharova et al. characterized the spectral and topographical signature of this contamination directly, showing that temporalis contraction produces EMG with peak power at 40–80 Hz but measurable energy from 0 to over 200 Hz, that this activity is strongest at temporal and adjacent electrodes, and that even weak (15% of maximum) contraction is sufficient to contaminate frequencies as low as 8 Hz [23]. Because EMG’s beta-range peak power has been reported as low as 13–38 Hz at frontal and temporal sites specifically [23], this contamination risk is not confined to high-gamma or the raw broadband signal alone; it plausibly extends to alpha- and beta-band features at AF7, AF8, TP9, and TP10 as well. Using a fundamentally different validation method—recording scalp EEG before and after complete pharmacological neuromuscular paralysis—Whitham et al. confirmed the magnitude of this effect directly rather than inferentially, finding 10- to 200-fold reductions in power above 20 Hz once muscle activity was eliminated, with the largest reductions at electrodes closest to temporal and cranial muscles [24]. Critically, this contamination is not contingent on deliberate muscle activity: using intramuscular single-motor-unit recordings simultaneous with EEG, Yilmaz et al. demonstrated that temporalis (and frontalis) motor

units discharge tonically even when participants are instructed to rest, that this tonic activity produces detectable interference potentials on nearby EEG electrodes, and that this interference is significantly reduced but never fully eliminated by independent component analysis (ICA), the standard artifact-correction technique [25]. This has a direct bearing on any interpretation of the broadband, unfiltered EEG signal specifically: because the undecomposed raw signal is not restricted to a conventional frequency band, it is structurally the signal component most exposed to this failure mode, since EMG’s broadband character means no low-pass or band-limiting step can cleanly separate it from genuinely neural activity. The present thesis does not apply ICA or other artifact-correction pipelines to the raw broadband signal (Section 3.6), and does not deploy dedicated EMG reference channels of the kind used in more recent correction approaches [?]. Consequently, whenever the raw broadband signal or a temporal-site (TP9/TP10) feature emerges as strongly discriminative between **STAY** and **SKIP** (Section 4.4.1), this literature motivates treating that result as ambiguous between a genuine neural signature and an artifact of jaw or temporalis muscle tension covarying with the skip decision, rather than treating it as evidence of one or the other by default. This ambiguity is addressed only descriptively in the present work (Section 4.5) and is flagged explicitly as a target for future artifact-controlled replication rather than resolved here.

2.12 Delta-Band Function in Awake Cognition

Although delta (0.5–4 Hz) oscillations are most extensively characterized during sleep, a separate and substantial literature establishes that delta oscillatory responses also play a functional role in awake human cognition. Güntekin and Başar’s review of evoked and event-related delta responses concludes that delta activity is consistently involved in decision-making and attentional processes across a wide range of cognitive paradigms, and that reduced delta responses during cognitive tasks are a candidate general marker of impaired cognitive processing, observed across Alzheimer’s disease, mild cognitive impairment, schizophrenia, bipolar disorder, and alcoholism [26]. From a complementary evolutionary-functional perspective, Knyazev argues that delta oscillations index basic attentional and motivational salience-detection processes that are evolutionarily older than the higher-frequency (theta/alpha/beta) systems typically associated with more advanced, flexible cognitive control; critically, this account predicts that delta-linked processing becomes functionally more prominent precisely when these higher-frequency systems are less dominant or less effective [27]. Taken together, this literature establishes

two points relevant to the present work: first, that delta activity is not merely a sleep or pathology marker but a recognized correlate of decision-making and attention in the awake brain; and second, that delta’s relative importance is expected to increase under conditions where composite, higher-frequency-weighted indices of engagement – such as the classical Engagement Index, $\beta/(\alpha + \theta)$ – fail to fully capture the relevant cognitive state. Neither line of work, however, has previously been applied to short-form video engagement or to a naturalistic, self-paced swiping paradigm; the only directly comparable evidence comes from short-form-video-addiction (SVA) research and is mixed rather than confirmatory. Xu et al., studying adolescents (mean age 12.3) during passive short-form video viewing, found reduced task-related delta engagement predicted worse working memory and cognitive control, though the effect was small and borderline-significant ($F=2.184$, $p=.040$, $R^2=.115$) [28]. Rahman et al., by contrast, found beta and gamma bands, not delta, to be the strongest predictors, though in a smaller sample ($n=10$) performing a coarser binary phone-use classification task (not a skip decision), using different gamma-band boundaries (30–44 Hz) than adopted here [4]. Neither paradigm resembles the self-paced STAY/SKIP decision studied in this thesis, which is itself evidence that the present question has not previously been addressed rather than that prior evidence on it is merely mixed. This thesis therefore carries forward a specific, literature-motivated but non-committal assumption: that delta activity is a plausible candidate for carrying decision- and attention-relevant variance in the present naturalistic browsing paradigm, motivated by its established general role in awake cognitive attention and salience detection rather than by prior TikTok- or SVA-specific validation, which remains unsettled. Section 4.4.1 tests this candidacy directly by evaluating delta-band separability against the Engagement Index and against other frequency bands, and Section 4.5 returns to this literature in interpreting the resulting AF8 delta peak-frequency effect.

2.12.1 Notch Filtering and the High-Gamma Band

Standard practice for removing power-line interference from EEG is to apply a narrowband notch filter centered on the local mains frequency (50 Hz in most of Europe). This practice is uncontroversial when the frequency range of interest lies outside the notch, but becomes a substantive methodological choice when it does not: the conventional high-gamma band (40–60 Hz) straddles 50 Hz directly, meaning any notch filter wide enough to reliably suppress line noise necessarily attenuates a portion of the high-gamma spectrum alongside it. This is not merely a hypothetical concern for low-cost, dry-electrode systems specifically. Pattisapu and Ray, comparing

a consumer-grade open-hardware amplifier (OpenBCI) against a research-grade system under identical conditions, found that the low-cost device exhibited pronounced mains-noise contamination together with additional intermodulation artifacts at frequencies flanking the line frequency (36 Hz and 64 Hz, symmetric around 50 Hz), an effect attributed to non-linear distortion introduced during amplification and not present in the research-grade comparison system [29]. It is worth noting this risk is not necessarily one-directional: the same comparison also found the consumer device to underestimate true gamma-band amplitude relative to the research-grade reference under matched conditions [29], indicating hardware-related gamma distortion in this device class is not uniformly an inflationary artifact. Critically, this contamination pattern falls inside the same 40–60 Hz range this thesis treats as high-gamma, meaning that for consumer EEG hardware of the kind used here, the region a notch filter would need to suppress and the region carrying putative cognitive high-gamma signal are not cleanly separable in frequency. No published evidence, to our knowledge, quantifies how much genuine high-frequency cognitive variance a standard notch filter removes as a side effect in this hardware class, nor conversely how much of an un-notched high-gamma signal reflects line-noise and intermodulation artifact rather than cortical activity; the literature establishes that the risk exists in both directions without resolving it. Given this, this thesis treats the notched-versus-un-notched decision as an open methodological trade-off rather than a settled default, and elects to preserve the unfiltered signal for all reported analyses so that no frequency content is discarded on an unverified assumption (Section 3.6); the resulting risk that reported high-gamma and broadband findings partially reflect line-noise contamination is carried forward explicitly as a limitation in the interpretation of those results (Section 4.5).

2.12.2 Pre-Motor Window and the Limits of the BP Literature

The choice of a pre-swipe gap wide enough to separate a decision-related signal from the motor act that follows it rests on two distinct bodies of evidence, only one of which speaks directly to motor contamination. The first is the Bereitschaftspotential (BP) literature: a slow cortical negativity precedes self-paced voluntary movement, with an early component beginning approximately 1.5 s before onset and a steeper late component beginning 400–500 ms before onset [16]. This literature is well established but answers a narrower question than it is often taken to answer – it characterizes when neural *preparation* for movement begins, not when the physical motor act

itself, or its peripheral muscular correlate, begins. A gap defined solely by BP onset times therefore establishes that a window lies outside the window of neural movement preparation; it does not by itself establish that the window is free of motor execution or muscular artifact, since the BP and the muscular event it precedes are not the same thing and are not interchangeable justifications for a single design choice. Lew et al. close part of this gap directly: in a self-paced reaching paradigm structurally comparable to the present swipe task, arm muscular (EMG) activation was measured, rather than inferred from BP timing, and found to begin on average 263 ± 40 ms before movement onset [15]. This is the only source in the present review that reports a directly measured, rather than literature-inferred, onset for the peripheral motor event itself. Taken together, these two findings support – but do not by themselves prove – the intuition motivating the present design: a gap of several hundred milliseconds to roughly 1.5 s should place a labeled window beyond both the neural preparatory signal (BP) and the measured onset of physical motor execution (EMG) in comparable self-paced tasks. This thesis adopts that intuition as the literature-motivated starting point for the 2.0-second pre-swipe gap (Section 3.4.1), while treating it explicitly as a margin derived from directly comparable but not identical paradigms – reaching versus swiping, arm versus thumb – rather than as a guarantee that the resulting **SKIP** window is free of motor-related signal. The task-specific validity of this assumption is not established by prior literature and is not tested independently in the present work; it is a boundary condition carried forward into the parameter sensitivity analysis (Section 3.4.4) rather than a settled fact.

Chapter 3

Materials and Methods

3.1 Participants

Twenty-eight ($N = 28$) participants were recruited for the study (14 female, 14 male). To participate, individuals were required to have normal or corrected-to-normal vision and no history of neurological disorders. Three participants were excluded prior to analysis: two due to a high number of self-reported keypress errors (greater than 10) which compromised label integrity, and one due to an undisclosed neurological condition. The experimenter’s own data were recorded separately for pipeline validation and were never part of the recruited sample; they are excluded from all reported results. The final sample consisted of $n = 25$ participants (age range 20–30 years, $M = 24.6$, $SD = 2.9$, 13 female, 12 male).

Participants were collected over a total of five days and were conceptually divided into two groups based on recruitment timing: an initial smaller group was recorded in a separate room without financial compensation, while the subsequent main group (recorded a few days later) received a 20-euro voucher as an incentive - this may be noted as a potential limitation of this study, and subject of further investigation beyond the scope of this work. All participants provided informed consent and signed data protection agreements approved by the Ethics Committee of the University of Potsdam prior to the study (No. 5/147, 08 Sep 2025; “Optimizing Large Scale Public Health Interventions on Social Media by Creating Neurological Markers of Engagement and Addiction in TikTok to Drive Disease Prevention”). Because recruitment relied substantially on university mailing lists, the sample skews toward an academically-oriented population; the implications of this for generalizability are addressed in Section 4.5 and in Future Research.

ID	Sex	Age	Daily SFV Use	Sleep (h)	Alertness	Consumed	Status
P4	F	26	1-30m	6-7	Neutral	Caffeine	Valid
P5	F	30	1-2h	7-8	Alert	None	Valid
P6	F	24	1-30m	7-8	Neutral	None	Valid
P7	F	23	30-60m	6-7	Neutral	None	Valid
P8	F	23	30-60m	6-7	Neutral	None	Valid
P9	M	27	30-60m	6-7	Neutral	Caffeine	Valid
P10	F	24	1-30m	>8	Neutral	Caffeine	Valid
P11	F	26	30-60m	7-8	Neutral	Caffeine	Valid
P12	F	20	30-60m	5-6	Neutral	Energy Dr.	Valid
P13	M	28	1-30m	>8	Neutral	Caffeine	Valid
P14	F	22	1-2h	>8	Tired	None	Valid
P15	M	28	1-30m	7-8, >8	Neutral	Caffeine	Valid
P16	M	32	1-30m	7-8	Alert	None	Excluded
P17	F	30	30-60m	6-7	Neutral	Caffeine	Valid
P18	M	27	1-2h	7-8	Alert	Caffeine	Valid
P19	F	30	1-30m	7-8	Neutral	Caffeine	Excluded
P20	M	20	1-30m	>8	Alert	Caffeine	Valid
P21	M	21	30-60m	6-7	Neutral	None	Valid
P22	F	21	1-30m	>8	Alert, Neutral	None	Valid
P23	M	23	1-2h	>8	Neutral	None	Valid
P24	F	24	1-2h	>8	Neutral	Caff+Nic	Valid
P25	F	21	>3h	>8	Neutral	None	Valid
P26	M	24	30-60m	6-7	Neutral, Tired	None	Valid
P27	M	23	2-3h	7-8	Alert	None	Valid
P28	M	28	30-60m	7-8	Very alert	None	Valid
P29	M	22	1-2h	>8	Very alert	None	Excluded
P30	M	26	>3h	5-6, 6-7	Very alert	None	Valid
P31	M	27	1-2h	6-7	Alert	None	Valid

Table 3.1: Participant demographics and pre-session survey responses. SFV = Short-Form Video. P16, P19, and P29 were excluded from the final analysis.

3.2 Data Acquisition

Electroencephalography (EEG) data were recorded using a consumer-grade Muse S headband (Generation 2, see Figure 3.1), which has been used successfully in similar EEG-based tasks [7, 30, 31]. The device features four dry electrodes placed at positions AF7, AF8, TP9, and TP10 according to the international 10-20 system, with the reference electrode located at Fpz. Two physical units of the same model were used across the cohort; device-related effects therefore cannot be fully separated from participant-related effects (see Section 4.4.1). Data were continuously sampled at a rate of 256 Hz, consistent with the hardware specifications of the Muse S headband [4].



Figure 3.1: Marketing material of the consumer-grade Muse S (Gen 2) EEG headband that was used for data acquisition. Source: [32]

The EEG signals were transmitted via Bluetooth and recorded through the Lab Streaming Layer (LSL) protocol using the `muselsl` library [5]. A custom Python recording script was developed to handle data collection (https://github.com/coffeeandcoffee/eeg_tiktok_pipeline). This script featured real-time frequency monitoring and an automatic reconnection mechanism to seamlessly manage any Bluetooth drops without disrupting the session.

To synchronize the behavioral data (video transitions) with the continuous EEG stream without relying on video monitoring of the participant, keystroke markers were mapped directly into the recording script. Participants used their non-dominant hand to press a specific key on a laptop keyboard provided in the room: the “B” key to denote the start of the baseline period, and the “A” key to mark every manual swipe (skip event) during the TikTok browsing session. The recording cabin and the physical arrangement of phone, laptop, and headband are shown in Figures 3.2 and 3.3.



Figure 3.2: The experimental setup cabin ensuring privacy and a distraction-free environment for the participant.



Figure 3.3: Close-up schematic sketch of the recording setup showing the researcher’s phone and the laptop used for keypress synchronization by the participant wearing the EEG headband while naturalistic TikTok browsing.

3.3 Experimental Protocol

The experimental sessions were scheduled for 45-minute timeslots, with the actual experimental procedures taking approximately 35 to 40 minutes. Upon arrival, participants were greeted, provided with an information sheet, and asked to sign the ethics consent forms. Afterward, they completed a pre-

experiment demographic and basic screening survey, which took approximately five to ten minutes.

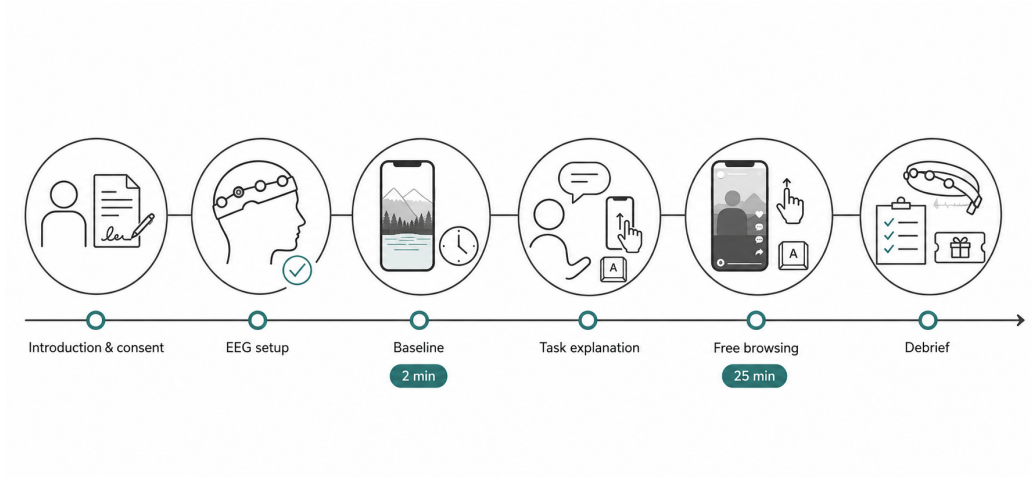


Figure 3.4: Experimental protocol flow: Introduction and consent, EEG setup, 2-minute baseline video, task explanation, and 25-minute TikTok free browsing session, debrief.

Figure 3.4 summarizes the resulting session structure. Participants were then instructed on the tasks. The Muse S headband was mounted on their head and adjusted to ensure optimal skin contact, particularly on the forehead and behind the ears for participants with long hair. The experimenter verified the signal quality and ensured the device was adequately charged.

The experimental paradigm consisted of two main phases: a baseline recording and a free-browsing session. For the baseline phase, participants were provided with the researcher’s iPhone, and a two-minute nature video on YouTube was started. Participants were instructed to relax and simply watch the video. Concurrently, the recording script was initiated, and the “B” marker was pressed. To ensure ecological validity while keeping the participant focused, the iPhone’s “Guided Access” feature was enabled to prevent exiting the application. Participants were left alone in the room to afford them privacy. After two minutes, the Guided Access lock visually timed out, signaling the conclusion of the baseline period.

The experimenter then returned to explain the second phase: a naturalistic short-form video browsing task using the TikTok application. To ensure a standardized yet personalized starting state, the app’s “For You” feed was refreshed using the “Reset feed” option in the content preferences settings.

Participants were instructed to consume TikTok content as they normally would, watching videos they found interesting and swiping away those they found boring. To restrict the action space strictly to scrolling behavior, they were asked to refrain from using the “like” function, writing comments, or visiting user profiles.

Crucially, participants were tasked with pressing the “A” button simultaneously with every swipe motion on the phone to accurately synchronize their micro-decisions with the EEG data. For instance, a participant holding the phone in their right hand used their left hand to press the designated key on the adjacent laptop keyboard; Figure 3.5 illustrates this synchronization step.

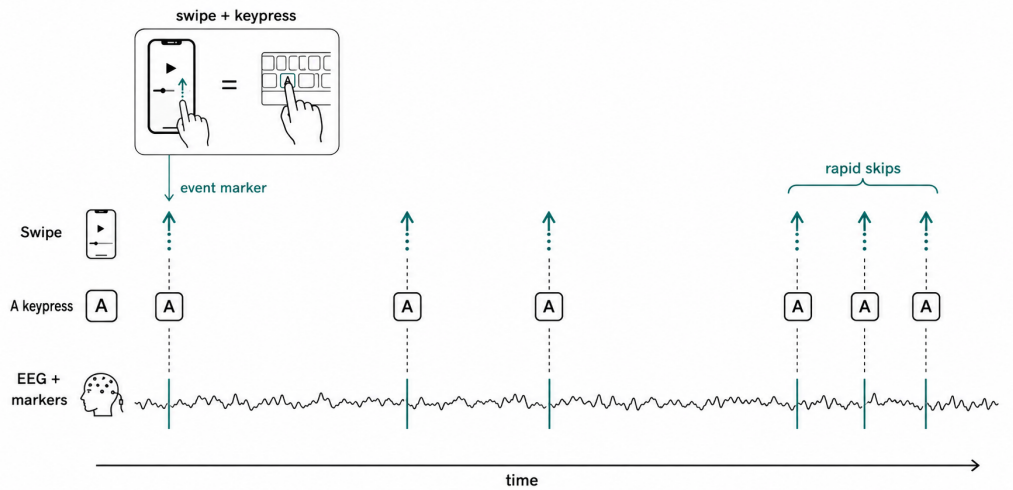


Figure 3.5: Schematic representation of the manual synchronization process: the participant swipes to skip the video and simultaneously presses the ‘A’ button to mark the event in the EEG stream.

The Guided Access feature was reactivated and set to a 25-minute limit for the TikTok app. Participants were reminded that the headband script would automatically reconnect if a Bluetooth disconnection occurred. They were then left alone in the room with the door closed, creating an isolated viewing environment designed to elicit natural responses to the video feed. They were instructed to knock on the glass door if any errors or issues arose; however, no significant interruptions occurred during the sessions.

Once the 25-minute viewing task concluded, the experimenter entered the room, and the participant removed the headband. They then answered a few post-session survey questions, specifically self-reporting if they made

any keypress synchronization errors during the session. Participants in the compensated group then signed the required forms to receive their 20-euro voucher. The post-measurement debrief and surveying process took roughly two to four minutes. Participants were thanked for their time, briefly debriefed about the broader context of the study if interested, and dismissed.

3.4 Labeling Strategy

To quantify class separability under different labeling choices, Cohen’s d was computed as the standardized mean difference between the STAY and SKIP distributions. A one-at-a-time parameter sensitivity analysis was then performed in which gap length, window length, and imminent-skip period were each varied while the remaining two were held fixed (Section 3.4.4).

3.4.1 Operational Definition of Labeling Parameters

The labeling strategy was derived directly from the research questions and the aim of isolating neural states relevant to sustained engagement and imminent disengagement during naturalistic browsing. Accordingly, the computational target was constrained to two discrete classes: sustained engagement (STAY) and impending disengagement (SKIP), reflecting the distinction between sustained attentional maintenance, understood as a fluctuating internal state rather than a fixed behavioral duration [1], and passive or dissociative browsing states in which continuous consumption need not indicate engagement [2, 3].

The default labeling scheme was defined by three temporal parameters. The gap length was set to 2.0 seconds, the window length was set to 1.0 second, and the imminent-skip period was set to 3.0 seconds; a fixed 0.5-second post-skip orientation grace period was additionally excluded from both classes. These defaults were not chosen arbitrarily: the 1.0-second window matches the value found to maximize classification performance in a direct 1–5 s window-length sweep on wearable EEG, where accuracy declined monotonically as window length increased [7]. The 2.0-second gap exceeds the onset of the early Bereitschaftspotential, reported at approximately 1.5 s before self-paced movement with a late, steeper component beginning 400–500 ms before onset [15, 16], so the gap was set beyond this window with a margin intended to reduce contamination of the SKIP epochs by motor-preparatory activity. This margin is further supported by directly measured EMG onset latency in a comparable self-paced reaching paradigm, where arm muscular activation began on average 263 ± 40 ms before movement

onset [15], placing the 2.0-second gap well beyond both the neural (BP) and peripheral (EMG) onset windows reported in that study. The 3.0-second imminent-skip period was not derived from a specific literature value and is treated as an heuristic starting point, addressed directly by the parameter sensitivity analysis in Section 3.4.4. In the default configuration, the **SKIP** class therefore comprised the 1.0-second windows tiled across the 3.0-second imminent-skip period preceding the 2.0-second gap before each registered **A**-keypress (swipe event), yielding a label placement that aligned with the pre-skip attentional state while avoiding the most immediate motor-preparatory interval. The **STAY** class comprised all remaining continuous viewing segments that satisfied the exclusion criteria described below.

3.4.2 Operational Definition of **STAY** and **SKIP**

The **SKIP** class was drawn from the imminent-skip period, defined as the 3.0-second interval terminating 2.0 seconds before the timestamp of each registered **A**-keypress (swipe event); the labeled **SKIP** epochs are the 1.0-second windows tiled within this interval. This interval captures neural activity preceding the physical act of skipping a video while placing the **SKIP** window beyond the motor-preparation interval, also known as Motor-Readiness-Potential or Bereitschaftspotential, which has been reported to occur approximately 1.5 s - 400 ms before movement onset [15]; the 2.0-second gap preceding the swipe marker therefore exceeds this interval, with limited margin at the earliest reported BP onsets. This gap also exceeds the directly measured onset of arm muscular (EMG) activation in a comparable self-paced task, reported at 263 ± 40 ms before movement onset [15], indicating that the gap is unlikely to be contaminated by the physical motor execution itself, though residual overlap with early BP onset cannot be fully excluded given between-subject variability.

The **STAY** class represents continuous, non-skip viewing periods, which are treated as a behavioral proxy for sustained engagement. It is noted that continuous viewing time is an imperfect proxy: passive dwell time can also reflect disengaged or dissociative states rather than effortful attention [2, 3], and sustained attention itself is better understood as a fluctuating internal state than a fixed behavioral duration [1]. No EEG marker has been shown to positively validate duration-based labeling as a proxy for cognitive engagement; recent work directly comparing sustained-attention and engagement spectral markers found engagement-sensitive markers failed to discriminate task conditions and that duration-aggregated metrics obscure real-time fluctuations in cognitive state [6], though the authors themselves attribute the frontal-midline-theta null result partly to a baseline confound and a mod-

est sample ($n=15$) rather than to an unqualified failure of the marker. The **STAY** class was defined by extracting all remaining continuous viewing periods during the naturalistic TikTok consumption phases that were i) outside the 3.0-second imminent-skip period, ii) outside the 2-second pre-skip motor-preparation buffer, and iii) outside the 0.5-second post-skip orientation grace period.

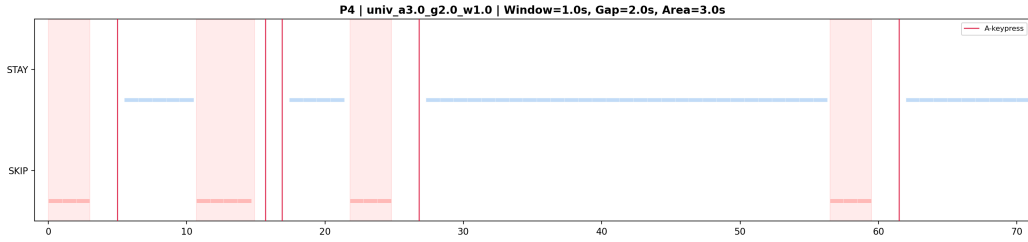


Figure 3.6: Schematic overview of the labeling strategy under the default parameterization. The figure illustrates the placement of the **STAY** and **SKIP** labels relative to the registered press markers, including the default gap length, window length, and imminent-skip period.

3.4.3 Baseline Exclusion

The manually categorized pre-task resting baseline phases were excluded from the **STAY**/**SKIP** classification dataset and used only for per-participant signal normalization. Consequently, the algorithms were trained to discriminate between continuous non-skip viewing (**STAY**) and the period preceding a swipe-away event (**SKIP**) during browsing, rather than between resting and viewing. Figure 3.6 illustrates the default labeling logic, including the press markers and the temporal placement of the labeling windows.

In addition to baseline exclusion, the labeling procedure omitted segments affected by boundary overlap between **STAY** and **SKIP** definitions to preserve clear class assignment and avoid ambiguous temporal regions near each swipe event, following established practice of excluding windows that do not lie entirely within a single labeled interval [7]. Boundary-adjacent mislabelling is known to be more probable and more consequential than uniformly distributed label noise [13], motivating this exclusion despite the resulting reduction in sample count. The final label construction therefore prioritized temporal purity of the target classes over maximal sample count.

3.4.4 Parameter Sensitivity Analysis

To assess how label definition affects class separability, a controlled one-at-a-time (OAT) sensitivity analysis [14, §2.4.2] over three temporal parameters was conducted: in each experiment two parameters were held constant while the third was varied, so that any change in separability is attributable to the varied parameter alone. This follows established practice in EEG decoding, where window length [7] and label latency [7] have been varied in isolation to quantify their effect on classification performance. Two limitations follow from the design. First, attribution is local to the operating point at which the remaining parameters were fixed [14]. Second, interaction effects are not identifiable under an OAT scheme [9]; resolving them would require a factorial sweep of the kind used for time window and frequency band in motor-imagery decoding [8]. Independent evidence for monotonic window-length gains comes from Tzamourta et al., who found EEG-classification accuracy increased step-wise from 5 s to 12 s windows across six classification problems and five brain regions with no exceptions [10]. Similarly, in MI-BCI paradigms, baselines positioned further from the upcoming task cue yielded significantly stronger ERD and higher classification accuracy than baselines positioned closer to the cue [11], mirroring the present gap-length and imminent-skip-period sensitivity. However, window-length effects are not always monotonic: a multi-scale CSP study found subject-specific optimal window sizes with accuracy differences of up to 13% depending on window choice [12], reinforcing that the present sweep should be read as suggestive rather than as evidence of a universal optimum.

The influence of window size (ws), gap size (gs), and imminent-skip period (isp) on data preprocessing and segmentation is illustrated in three example-configurations: (i) default parameters (ws = 1s, gs = 2s, isp = 3s; Figure 3.6), (ii) reduced window size (ws = 0.3s, gs = 2s, isp = 3s; Figure 3.7), and (iii) reduced gap with extended imminent-skip period (ws = 1s, gs = 0.5s, isp = 4s; Figure 3.8). The swipe event is indicated by vertical red lines. All figures show the same randomly selected 70-second segment from a single participant. Time (seconds) is plotted on the x-axis, and class labels—STAY (blue) and SKIP (red)—on the y-axis.

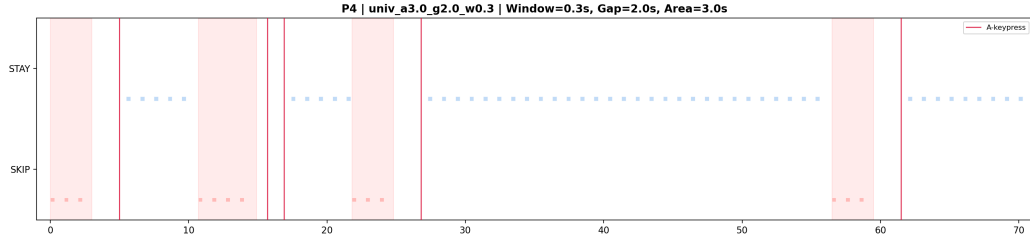


Figure 3.7: Window parameter exploration (default parameters, except a smaller window size).

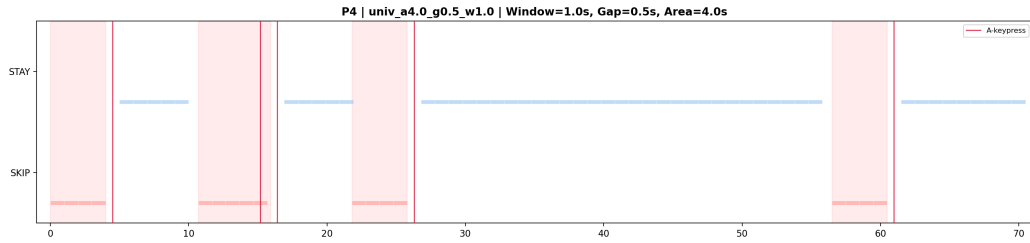


Figure 3.8: Window parameter exploration (default parameters, except a smaller gap size & a longer imminent-skip period (Skip-Area)).

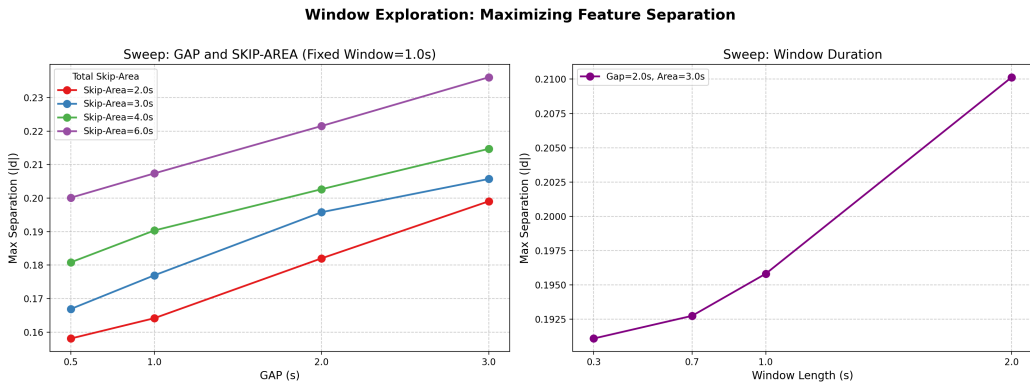


Figure 3.9: Parameter sweep of the window exploration analysis. This figure originates from an earlier pipeline run predating a correction to the Engagement Index computation; the sweep evaluates only individual component features and does not use the Engagement Index, so the correction does not affect the values shown.

The key result is that, with fixed imminent-skip period and window size, increasing the gap length consistently increases Cohen’s d (Figure 3.9, left), indicating greater separability as the SKIP interval moves further from the swipe event. A similar effect is observed for longer imminent-skip periods, where larger SKIP intervals also improve separability. For fixed gap and imminent-skip period, increasing window size is monotonically associated with larger Cohen’s d (Figure 3.9, right), again corresponding to increased class separation. This pattern is consistent with independent findings in EEG classification more broadly: accuracy has been shown to increase monotonically with window length [10], and separability has been shown to increase with distance between the measurement window and the upcoming event [11]. At the same time, this monotonicity is not universal across EEG paradigms, as optimal window length has also been shown to be subject-specific rather than fixed [12]. While all these results strongly suggest better separability with increased overall data-availability (bigger windows, and areas) and simultaneously better separability the further we move away from the swipe, interpreting these relationships and their implications for optimal parameter selection remains beyond the scope of this thesis as it requires separate analysis, and is therefore a promising direction for future work. To maintain focus and scope, this analysis proceeds with the locked, default parameters:

- Window size (ws) = 1 second
- Gap size (gs) = 2 seconds
- Imminent-skip period (isp) = 3 seconds

3.4.5 Class Separation Metric

Cohen’s d is the standard scale-invariant index for quantifying the standardized difference between two distributions, expressing a mean difference in units of pooled standard deviation rather than in the arbitrary units of the underlying measure [17]. This property is what makes it suitable for comparing separability across heterogeneous feature types — spectral power, entropy, connectivity, or raw amplitude — that would otherwise be incommensurable on their native scales. The metric has established precedent specifically in EEG-based group and class separability analysis: Henao Isaza et al. use Cohen’s d to rank EEG-derived spectral, connectivity, and entropy features by their ability to separate Alzheimer’s-risk carriers from healthy controls, treating it as a principled feature-selection criterion prior to classifier training [18]. Kiarashi et al. apply the same metric to quantify group separability between MCI and cognitively normal participants directly from

model output distributions, and further show that separability under Cohen’s d responds systematically to methodological choices — in their case, multi-night signal aggregation — with (d) increasing alongside statistical significance as aggregation improved [19]. Taken together, this precedent establishes two points relevant to the present work: first, that Cohen’s d is an appropriate and validated tool for ranking discriminative EEG features independent of their native units, motivating its use here for feature-level class-separation ranking (Section 4.4.1); and second, that Cohen’s d is equally appropriate as a sensitivity metric for tracking how a labeling or preprocessing parameter choice affects downstream separability, motivating its use here for the labeling-parameter sensitivity analysis (Section 3.4.4). Neither prior use, however, involves a self-paced behavioral labeling scheme comparable to the STAY/SKIP paradigm applied here — extending this precedent to that specific setting is a contribution of the present work.

Cohen’s d expresses the standardized difference between the STAY and SKIP distributions, making it suitable for comparing separability across parameter settings even when the underlying distributions differ in scale. In this context, larger absolute values of Cohen’s d indicate stronger separation between the two classes.

$$d = \frac{\bar{x}_{\text{STAY}} - \bar{x}_{\text{SKIP}}}{\sqrt{\frac{(n_{\text{STAY}}-1)s_{\text{STAY}}^2 + (n_{\text{SKIP}}-1)s_{\text{SKIP}}^2}{n_{\text{STAY}} + n_{\text{SKIP}} - 2}}} \quad (3.1)$$

where $\bar{x}_{\text{STAY}}$ and $\bar{x}_{\text{SKIP}}$ denote the sample means of the feature values for the STAY and SKIP windows respectively, s_{STAY}^2 and s_{SKIP}^2 are the corresponding sample variances, and n_{STAY} and n_{SKIP} represent the number of samples in each class.

For each parameter configuration, Cohen’s d was computed independently between the STAY and SKIP distributions of every extracted univariate feature (e.g. `TP10_raw_std`), yielding one $|d|$ value per feature. This provided a direct quantitative index of how strongly the chosen temporal definitions emphasized the contrast between the STAY and SKIP behavioral classes. The two levels at which $|d|$ is reported in the feature-ranking analyses—exact per-feature values versus category means—are described in Section 3.8.2.

3.5 Universal Recording Pipeline

To facilitate rigorous and reproducible data collection, a robust Python-based recording architecture (https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_collection/recording_script.py) was developed. This

script orchestrates the simultaneous capture of raw EEG telemetry from the Muse S headband and has the capability of recording synchronized behavioral video from a webcam which was not used in this experiment, yet may be of value to future investigations. Its autonomous capabilities ensure that the experimental protocol can be executed reliably without extensive technical oversight.

For this study’s recordings the command `python recording_script.py --nocamera --duration 1800` was used to call the script. The two flags simply mean (i) bypassing camera recording, as it is not in the scope of this study yet kept for future investigations and (ii) the recording duration is 1800s = 30 minutes. The pipeline’s core features include:

- **Automated Hardware Discovery:** Upon launch, the script automatically scans the local Bluetooth environment for active Muse devices and enumerates all connected camera peripherals, allowing the researcher to select the optimal hardware interactively without requiring hardcoded MAC addresses or device indices.
- **Precision Synchronization and Event Logging:** A dedicated, threaded keypress listener (`pynput`) monitors the A and B keys used by participants to mark swipe events and baseline respectively. Crucially, the script computes a dynamic time offset between the system clock and the LSL stream clock. Keypresses are retroactively aligned to this LSL timeline within a defined tolerance window and embedded directly into the EEG data rows as binary flags, ensuring absolute temporal synchrony between cognitive events and physical interactions.
- **Simultaneous Multimodal Capture:** The script utilizes the `pyls1` library to interface with the Lab Streaming Layer (LSL), pulling raw EEG data from the Muse headband at 256 Hz. Unless deactivated with the flag, it concurrently employs OpenCV to capture and encode synchronized webcam video using the H.264 (avc1) codec, providing visual ground truth of participant behavior.
- **Resilient Auto-Reconnection:** To combat the inherently unstable nature of Bluetooth Low Energy (BLE) connections, the script features an active connection-loss detection loop. If the LSL stream halts (no samples received within a timeout period), the script automatically terminates the hanging subprocess, rescans the Bluetooth environment, restarts the stream, and initiates a sequential data fragment (e.g., `_2.csv`, `_3.csv`). This guarantees that transient connection drops do not corrupt the entire recording session.

- **Concurrent Multi-Threading:** To ensure that blocking operations (such as writing high-resolution video frames to disk) do not disrupt the high-frequency polling required for EEG telemetry, the script isolates the LSL pulling loop, the OpenCV capture loop, and the keypress listener into independent Python daemon threads.

This open-source compatible architecture ensures that any independent researcher equipped with a standard webcam and a consumer-grade Muse S headband can fully replicate the data collection methodology utilized in this thesis.

3.6 Universal Pre-Processing Pipeline

The following subsections describe step-by-step exactly what the Python code in the https://github.com/coffeeandcoffee/eeg_tiktok_pipeline directory executes to process the raw EEG signals into a uniform shape suitable for analysis.

3.6.1 Data Synchronization and Interpolation

The preprocessing pipeline began with the chronological alignment and concatenation of all recorded CSV files for each participant, ensuring a continuous temporal sequence (executed via https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_analysis/step01_preprocess.py). The raw data streams collected from the LSL (Lab Streaming Layer) exhibited slight variations in intrinsic sampling rates. To establish a uniform temporal grid suitable for across-session and across-participant analysis, the true sampling rate was derived by calculating the median time difference (Δt) between consecutive timestamps. Subsequently, a linear interpolation algorithm was applied to upsample and regularize the data to a constant sampling frequency of exactly $f_s = 256$ Hz. This process guarantees that any subsequent spectral analysis relies on a strictly uniform time axis, avoiding mathematical artifacts stemming from temporal jitter. Figure 3.10 confirms that all participants' signals operate at exactly 256 Hz after interpolation.

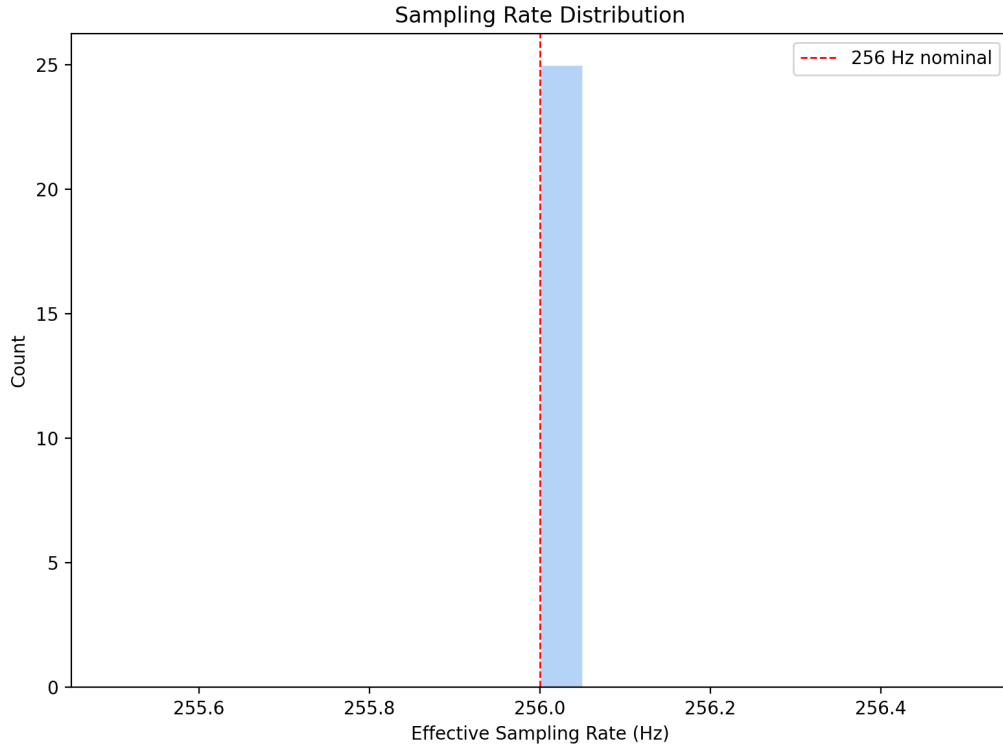


Figure 3.10: Sampling Rate Distribution across participants, demonstrating that after interpolation all signals operate precisely at 256 Hz, which is a mathematical prerequisite for the correct extraction of spectral band power.

During this temporal alignment process, Bluetooth transmission dropouts were identified by locating unnatural temporal gaps between consecutive CSV boundaries. To ensure data integrity, the duration of these dropouts was logged, and any potential data windows traversing such boundaries were strictly excluded in later extraction phases. Figure 3.11 shows the distribution of these events across the experimental timeline.

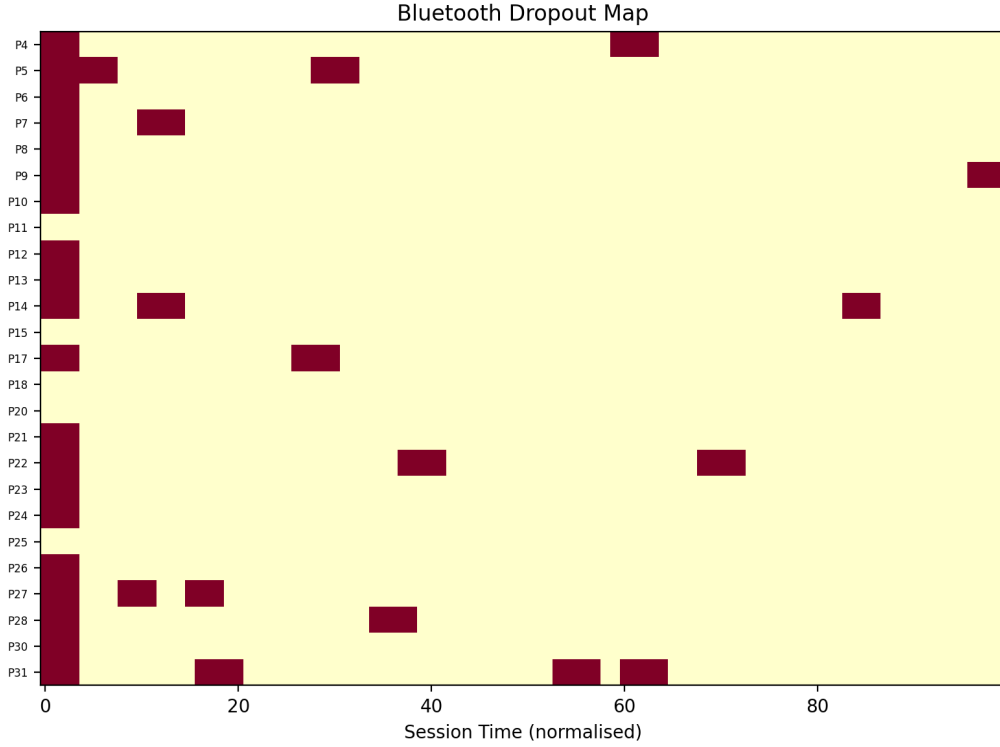


Figure 3.11: Bluetooth Dropout Heatmap indicating signal loss events across the experimental timeline. Data windows overlapping with these dropouts were removed from the analysis pipeline.

3.6.2 Baseline Extraction and Normalization

Following temporal synchronization, a baseline resting state was extracted for each participant to account for individual physiological differences, such as varying skull thickness or inherent resting cortical activity. The baseline period was rigidly defined as a 90-second continuous window commencing 10 seconds after the participant’s initial B keypress (executed via https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_analysis/step01_preprocess.py).

The interpolated signal was then separated into seven standard EEG frequency bands using a 4th-order Butterworth bandpass filter: Delta (1–4 Hz), Theta (4–8 Hz), Alpha (8–13 Hz), Beta (13–30 Hz), Low Gamma (30–40 Hz), High Gamma (40–60 Hz), and Very High Gamma (60–100 Hz). The undecomposed broadband signal was retained alongside these bands and

is referred to throughout as the *raw* signal. The instantaneous amplitude envelope A for each band was then obtained via the Hilbert transform. The linear envelope is used directly; it is not squared to power.

To normalize the extracted features across participants, a z-score normalization was applied against the participant-specific baseline. For every temporal sample, the normalized band amplitude z was computed as:

$$z = \frac{A - \mu_{base}}{\sigma_{base}} \quad (3.2)$$

where μ_{base} and σ_{base} denote the mean and standard deviation of the band amplitude envelope derived exclusively from the aforementioned 90-second baseline period. This normalization ensures that all subsequent statistical evaluations and machine learning models operate on normalized deviations from a participant’s own resting state, effectively mitigating absolute inter-subject voltage variance.

3.6.3 Windowing and Class Labeling

Following normalization, the continuous data streams were segmented and labeled into discrete **STAY** and **SKIP** classes based on the participant’s interaction events (**A** keypresses) as previously established in Sections 3.4.1 and 3.4.2. To strictly prevent motor artifact contamination arising from the physical swipe action, a **2.0-second gap** was enforced immediately preceding the **A** keypress, and a **0.5-second gap** was enforced immediately following it. No data was extracted from these gap periods.

The imminent-skip period was rigorously defined as the **3.0-second interval** immediately preceding the aforementioned 2.0-second pre-swipe gap, and all **SKIP** windows were drawn from within it (executed via https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_analysis/step03_label_windows.py). This temporal framing is intended to isolate the cognitive decision-making phase leading up to the swipe from the physical execution of the swipe itself; the 2.0-second gap exceeds both the reported onset of the Bereitschaftspotential (1.5 s–400 ms before movement) and the directly measured onset of arm muscular activation (263 ± 40 ms before movement) reported in a comparable self-paced reaching paradigm [15], though full elimination of motor-preparatory contamination cannot be guaranteed given individual variability in onset timing. The **STAY** class was defined as all remaining valid temporal periods outside of the **SKIP** windows and safety gaps.

Finally, discrete 1.0-second analysis windows were extracted using a sliding window with no overlap (stride = window length). This ensures that

windows do not cross class boundaries, enter safety gaps, or overlap with adjacent samples, thereby preventing data leakage and artificially inflated model performance, consistent with the practice of excluding windows that do not lie entirely within a single labeled interval [7].

3.6.4 Artifact Analysis

To ensure the integrity of the analyzed EEG signals, an automated outlier detection approach was implemented to flag potential artifacts (executed via https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_analysis/step02_artifact_flag.py). The preprocessing pipeline was executed twice, once without and once with a 50 Hz notch filter. The artifact-detection procedure described in this section relies on peak-to-peak amplitude thresholding, a coarse, broadband measure targeting large-amplitude deflections from blinks and muscle activity; for this purpose only, the notch-filtered variant was used so that a constant, non-physiological line-noise component would not itself inflate the amplitude threshold. All analyses reported elsewhere in this thesis, including the frequency-band and gamma-range results in Chapter ??, instead use the un-notched data: because the line frequency sits inside the conventional high-gamma range (40–60 Hz), notch filtering is not a neutral operation, and this thesis prioritizes preserving the full, unaltered frequency content over pre-emptively discarding a band on the assumption that its content near 50 Hz is purely artifactual. Specifically, the 95th percentile of the peak-to-peak amplitude computed on the notch-filtered signal was utilized as a dynamic, participant-specific threshold. Frontal channels (AF7, AF8) were evaluated for ocular artifacts (e.g., blinks), while temporal channels (TP9, TP10) were evaluated for electromyogenic (EMG) artifacts, consistent with prior evidence that temporal electrode sites are disproportionately susceptible to EMG contamination [23, 24]; this susceptibility has been reported at frequencies as low as 13 Hz [23], so it is treated here as a risk extending into alpha and beta bands, not only into high-gamma or broadband features. Absolute voltage thresholds (e.g., $150\ \mu\text{V}$) were initially investigated but were ultimately discarded as they did not yield generalizable or useful results across the dataset.

No data points were excluded from the final machine learning pipeline based on these artifact flags. Incorporating such filtering would introduce additional experimental conditions and substantially expand the analysis beyond the scope of this proof-of-concept study. Instead, the approach is documented to highlight awareness of artifact-related confounds and to provide a foundation for more advanced rejection strategies in future work. The resulting artifact-rate distribution is shown in Figure 3.12.

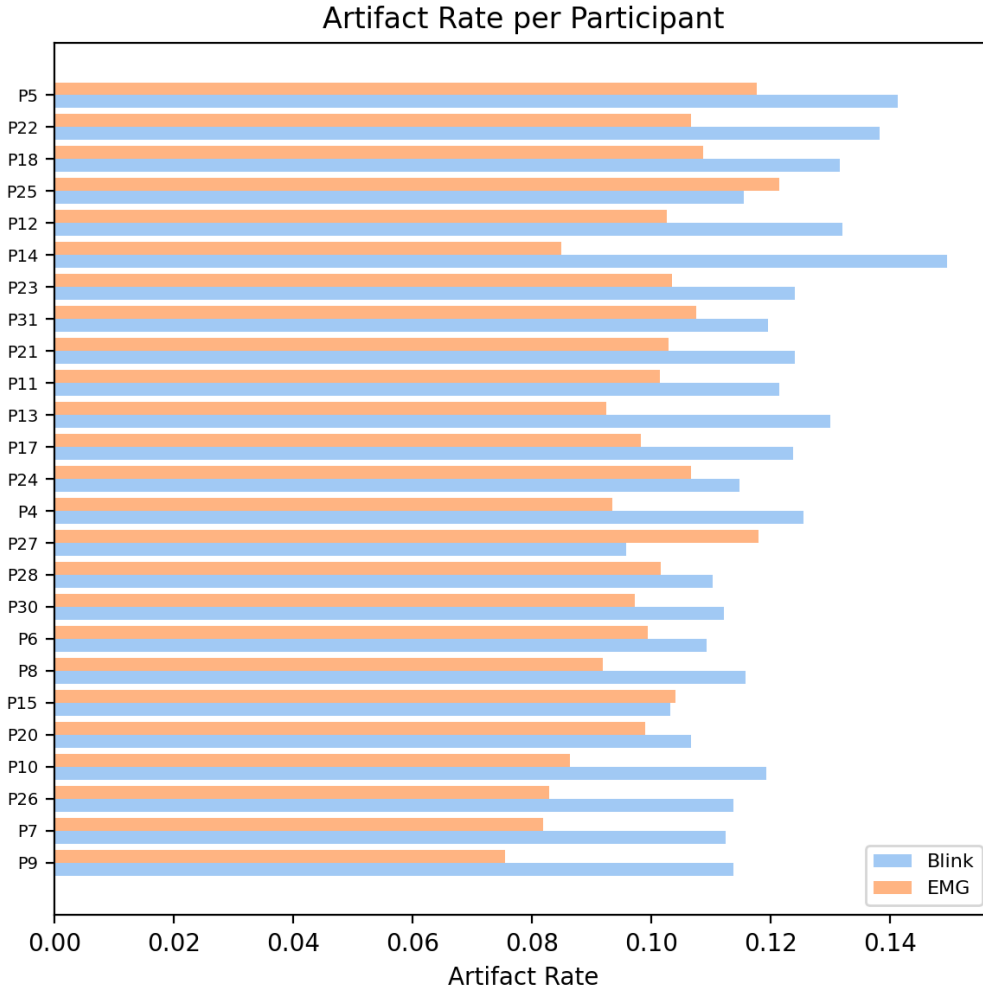


Figure 3.12: Artifact rate distribution and signal quality flagging based on the 95th percentile peak-to-peak outlier approach.

3.6.5 Behavioral Burst Analysis

In addition to physiological artifacts, behavioral anomalies termed “bursts” were identified. A burst is defined as a sequence of rapidly successive skip events (e.g., multiple A keypresses merging into a single contiguous temporal block), reflecting frantic swiping rather than deliberate evaluation. Flagging these events aims to ensure that the SKIP class captures deliberate pre-swipe evaluation periods rather than high-momentum, rapid-succession scrolling behavior.

Although bursts were successfully detected, they were not excluded from the final analysis. Figure 3.13 visualizes the detection mechanism via the distribution of inter-swipe intervals (i.e., dwell time) across all participants. The vertical threshold marks the 6.0-second merge boundary (twice the 3.0-second imminent-skip period). Consecutive swipes below this threshold cause their respective **SKIP** windows to overlap and merge into a single contiguous block, thereby defining a burst. The shaded region highlights the frequency of such rapid, consecutive events.

It is important to note that the imminent-skip period is defined independently of the dwell time distribution; the 3.0-second value is an intuitive choice. As such, its influence on the resulting burst structure warrants further investigation. Incorporating burst rejection would introduce additional analytical complexity and is therefore beyond the scope of this proof-of-concept study. Instead, its inclusion highlights a relevant behavioral confound and motivates future work.

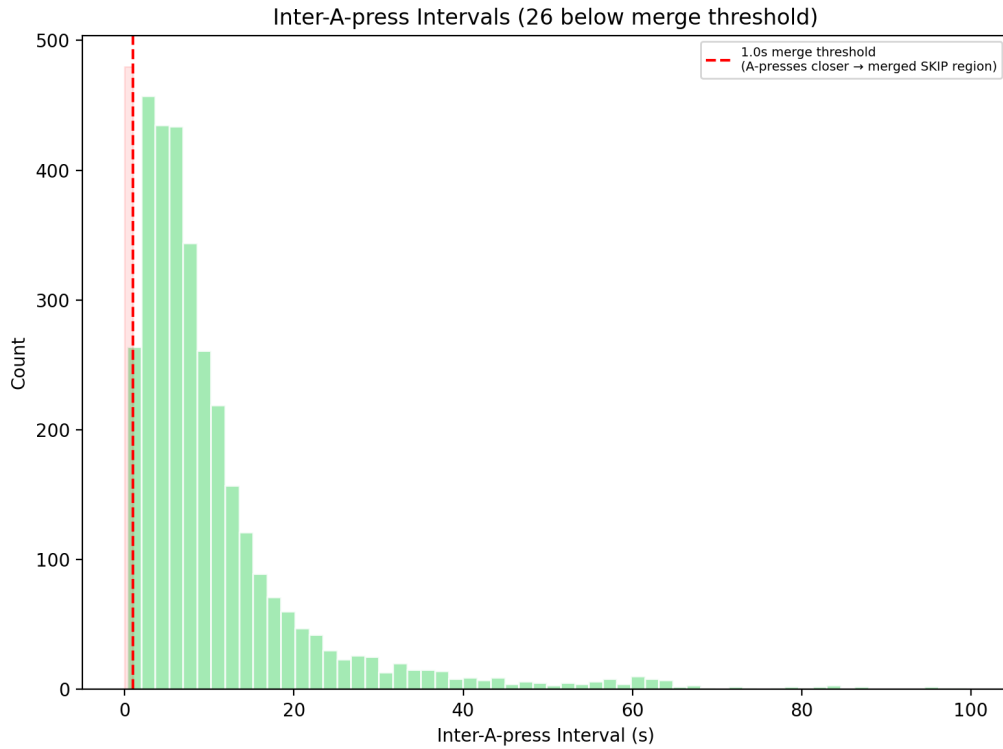


Figure 3.13: Distribution of inter-swipe intervals (view times of each video across all participants, or dwell time). The shaded region highlights intervals below the 6.0-second merge threshold, where consecutive **SKIP** windows overlap to form behavioral burst blocks.

3.7 Feature Engineering

3.7.1 Band Power and Statistical Moments

Following window extraction, a comprehensive feature set was computed for each valid temporal window (executed via https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_analysis/step05_feature_engineering.py). Two distinct extraction schemes are applied. For each of the seven frequency bands at each of the four electrodes, the mean and standard deviation of the normalized envelope are computed, alongside the relative power, Hjorth, and peak/macro frequency features described in the following subsections. Minimum and maximum were computed during development but are excluded from the final feature set. The undecomposed raw broadband signal

is treated separately and is described at the end of this section. An example of the extracted statistics for one 1-second window of one frequency band (alpha) at one electrode of one participant is depicted in Figure 3.14.

In addition to these four moments, a Relative Power feature was computed for each band and electrode as the mean power of that band divided by the summed mean power across all bands at the same electrode. This expresses each band’s contribution as a proportion of the total spectral power measured at that electrode and is therefore invariant to overall amplitude scaling.

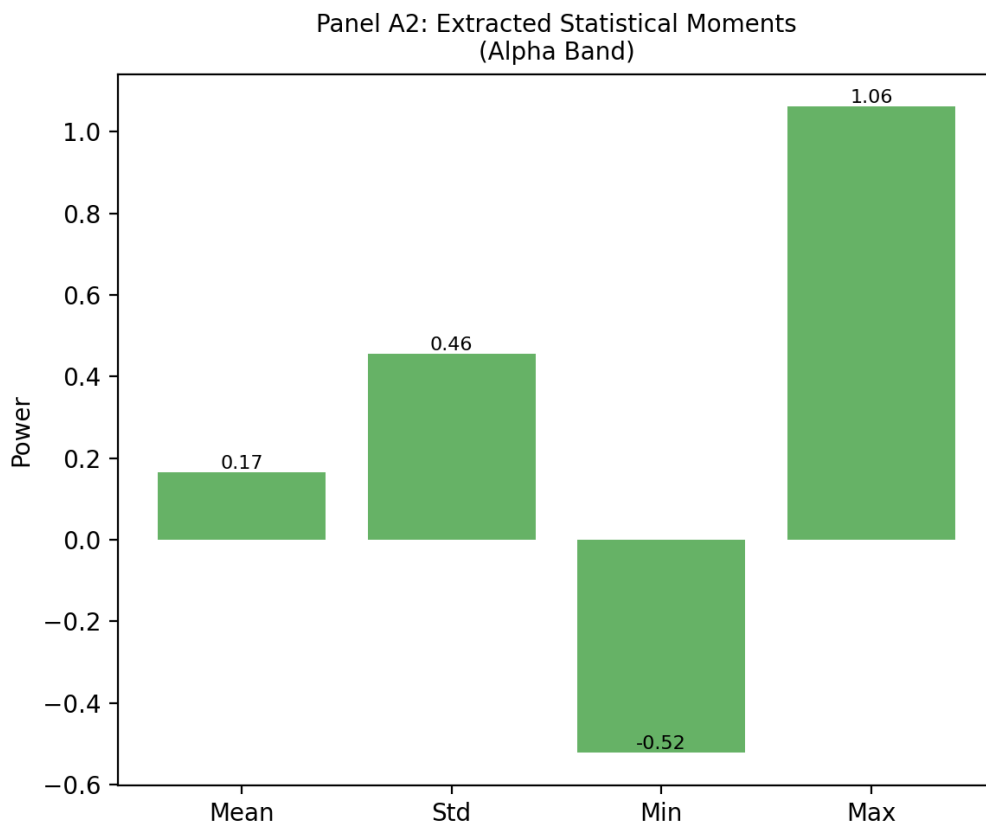


Figure 3.14: Extraction of standard statistical moments from the normalized EEG frequency bands.

3.7.2 Engagement Index Calculation

To provide a recognized clinical baseline for cognitive evaluation, the Engagement Index (EI) was calculated for each analysis window. The EI is

theoretically grounded in established neurophysiological research correlating task engagement with frontal and parietal band power ratios. It is mathematically defined as:

$$EI = \frac{\beta}{\alpha + \theta} \quad (3.3)$$

In this pipeline, exactly one scalar Engagement Index (EI) is computed per 1.0-second analysis window per participant. To calculate this single value, the within-window variance of the baseline-normalized band signal is first extracted for the Beta (β), Alpha (α), and Theta (θ) bands across each of the four individual electrodes. Because the normalized signal is z-scored and therefore centred near zero, the ratio is formed from variances rather than means; the variance is strictly non-negative, so the denominator cannot approach zero and the numerical instability of a mean-based ratio is avoided. These four spatial variances are then averaged together to produce a single, whole-head β , α , and θ scalar for that specific window, which are finally fed into the EI ratio formula. This single composite scalar serves as the primary clinical benchmark against which the predictive power of individual statistical features is evaluated in the machine learning phase. An example calculation of the EI for a single sample window of one participant can be observed in Figure 3.15, visually showing the three spatially-averaged scalars that are utilized in the Index calculation in Equation 3.3.

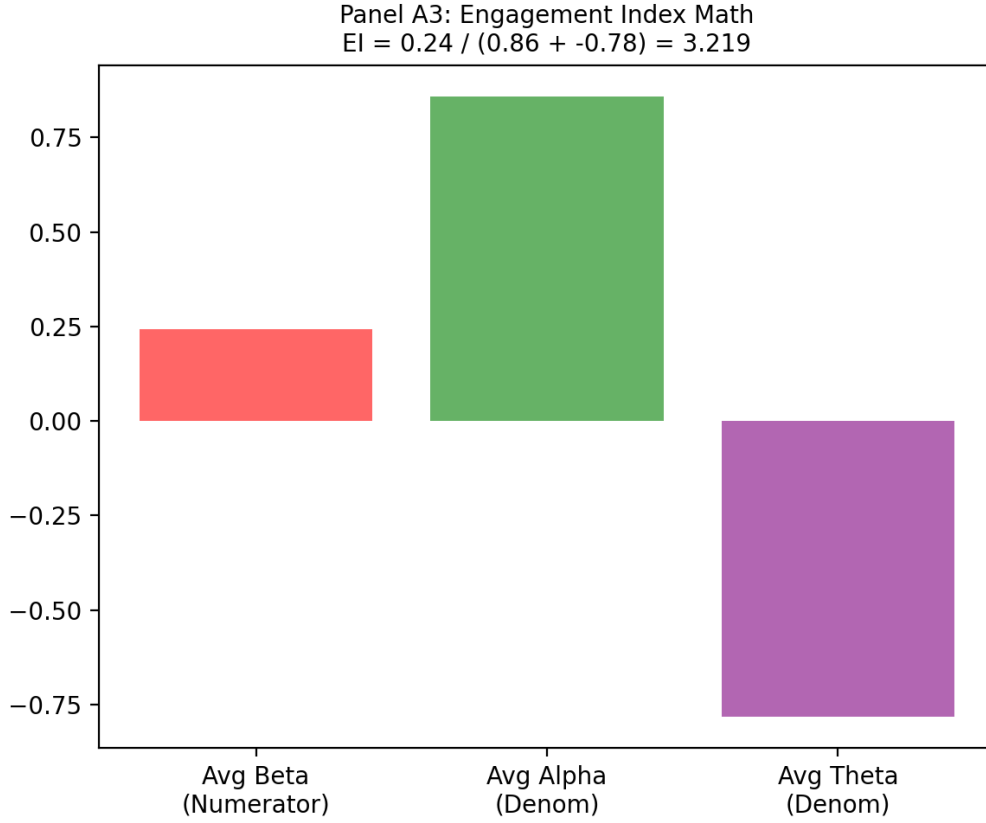


Figure 3.15: Engagement Index (EI) calculation deriving a single scalar engagement metric from the Beta, Alpha, and Theta band powers.

3.7.3 Hjorth Parameters

In addition to basic descriptive statistics, Hjorth Parameters (Activity, Mobility, and Complexity) were computed to capture the sequence-dependent temporal dynamics of the signal within the 1.0-second window. While standard statistical moments characterize the amplitude distribution independently of their temporal order (i.e., they are time-invariant), Hjorth parameters depend explicitly on the time-domain derivatives, thus capturing spectral properties—such as mean frequency and bandwidth—directly from the time-domain signal.

For a discrete EEG signal vector $y(t)$ of length $N = 256$ representing one frequency band at one channel within a window, the three parameters are mathematically defined as follows:

Activity

Activity measures the total signal power and is defined simply as the variance of the signal amplitude:

$$\text{Activity} = \sigma_y^2 = \frac{1}{N} \sum_{t=1}^N (y(t) - \mu_y)^2 \quad (3.4)$$

where μ_y is the mean of $y(t)$. The input dimension is the normalized amplitude and the output dimension represents the squared amplitude.

Mobility

Mobility serves as a time-domain proxy for the mean frequency of the power spectrum. It is defined as the square root of the ratio of the variance of the first derivative $y'(t)$ to the variance of the primary signal $y(t)$:

$$\text{Mobility} = \frac{\sigma_{y'}}{\sigma_y} = \sqrt{\frac{\text{var}(y'(t))}{\text{var}(y(t))}} \quad (3.5)$$

Since the differentiation of a signal with respect to time yields a rate of change, the resulting output dimension of Mobility is expressed in Hertz (Hz).

Complexity

Complexity measures the bandwidth of the signal and indicates how closely the signal's shape resembles a pure sine wave (where a pure sine wave converges to a complexity of 1). It is defined as the ratio of the Mobility of the first derivative $y'(t)$ to the Mobility of the primary signal $y(t)$:

$$\text{Complexity} = \frac{\text{Mobility}(y')}{\text{Mobility}(y)} = \frac{\sigma_{y''}/\sigma_{y'}}{\sigma_{y'}/\sigma_y} \quad (3.6)$$

where $\sigma_{y''}$ is the standard deviation of the second derivative of the signal. Because it is a ratio of two frequency-domain (Hz) equivalents, Complexity is a dimensionless scalar.

Together, these three parameters output exactly three scalar features per frequency band per channel, expanding the univariate feature space by explicitly encoding the temporal structural variations that basic amplitude statistics fundamentally miss.

3.7.4 Peak and Macro Frequency Extraction

Beyond traditional amplitude statistics, two specialized time-domain features were engineered to capture the dominant oscillatory and structural patterns within each frequency band of the 1.0-second analysis window. For a discrete band-pass filtered EEG signal vector $y(t)$ of length $N = 256$ and total duration $T = 1.0$ s, the features are calculated as follows:

Peak Frequency

Rather than extracting the maximum bin from a Fourier transform, Peak Frequency is calculated directly in the time domain by identifying the total number of local maxima (peaks) within the signal. A point $y(t)$ is considered a peak if it is strictly greater than its immediate neighbors. Let P be the set of all peak indices. The Peak Frequency f_{peak} is defined as the absolute peak count normalized by the window duration:

$$f_{peak} = \frac{|P|}{T} \quad (3.7)$$

The input dimension is the time-series amplitude vector, and the output is a scalar frequency representing the unsmoothed oscillatory rate of the frequency band in Hertz (Hz). Peak-counting is conceptually related to, but distinct from, established validated time-domain frequency estimators: Hjorth’s Mobility parameter derives a mean-frequency estimate from the variance ratio of a signal’s first derivative to the signal itself [20], and the zero-crossing rate used in speech and EEG classification pipelines derives a frequency proxy from the rate of sign changes [21]. Peak Frequency instead counts local maxima directly; this specific operationalization has no independent validation in the literature and is treated here as a heuristic descriptor rather than an established measure, addressed empirically via the ablation in Section 4.4.1.

Macro Frequency

While Peak Frequency captures every local maximum, it is highly sensitive to noise and high-frequency jitter. To isolate the true, underlying macroscopic oscillatory bursts, a novel Macro Frequency algorithm was developed.

First, a dynamic amplitude threshold θ_{peak} is calculated as the mean amplitude of all previously identified peaks:

$$\theta_{peak} = \frac{1}{|P|} \sum_{p \in P} y(p) \quad (3.8)$$

The signal is then converted into a binary sequence $B(t)$ where $B(t) = 1$ if $y(t) \geq \theta_{peak}$ and 0 otherwise.

Because a single macroscopic wave might contain slight amplitude dips that prematurely break a contiguous block of ones, the algorithm calculates the average length of all zero-sequences (gaps), denoted as $\bar{G}$. Any gap in $B(t)$ with a length $g \leq \bar{G}$ is mathematically bridged (converted to ones), producing a smoothed binary sequence $B_{smooth}(t)$.

Finally, the Macro Frequency f_{macro} is defined as the total number of distinct, contiguous blocks of ones in $B_{smooth}(t)$, normalized by the window duration:

$$f_{macro} = \frac{\text{BlockCount}(B_{smooth})}{T} \quad (3.9)$$

The output is a stabilized, macroscopic frequency in Hertz (Hz) that robustly captures the count of dominant structural waves within the cognitive window, effectively ignoring micro-fluctuations.

3.7.5 Raw Broadband Features

The undecomposed raw broadband signal is not passed through the band-specific feature set described above; instead, a separate set of time-domain descriptors is extracted directly from the raw signal. Treating the raw signal as a distinct feature class is motivated by evidence that broadband/aperiodic activity can carry decodable information independent of, and sometimes exceeding, band-limited oscillatory features [33, 34], and that band-power ratio measures can conflate periodic and aperiodic components [34]. Three time-domain descriptors are extracted per electrode: the mean, the standard deviation, and the linear slope of the signal across the window, the latter obtained as the gradient of a least-squares linear fit and expressed per second. Together these capture the amplitude level, the amplitude variability, and the direction and steepness of any sustained drift within the window.

3.7.6 Resulting Feature Space

Taken together, the two extraction schemes yield the final feature space. For each of the seven frequency bands at each of the four electrodes, eight statistics are computed—mean, standard deviation, relative power, the three Hjorth parameters, and the peak and macro frequencies—giving $7 \times 4 \times 8 = 224$ features. For the raw broadband signal, three time-domain statistics are extracted at each electrode, giving a further $1 \times 4 \times 3 = 12$ features. The complete feature space therefore comprises 236 univariate features per 1.0-second analysis window.

3.8 Result Synthesis Pipeline

3.8.1 Temporal Cross-Validation and Data Balancing

Before proceeding to statistical evaluation and modeling, the inherent class imbalance depicted in Figure 3.16, where the **STAY** class (blue) heavily dominates the **SKIP** class (red), had to be addressed (executed via https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_analysis/step04_balance_split.py). To prevent the models from trivially predicting the majority class, the **STAY** class was randomly undersampled for each participant to achieve a strict 50/50 balance, per participant, as depicted in Figure 3.17.

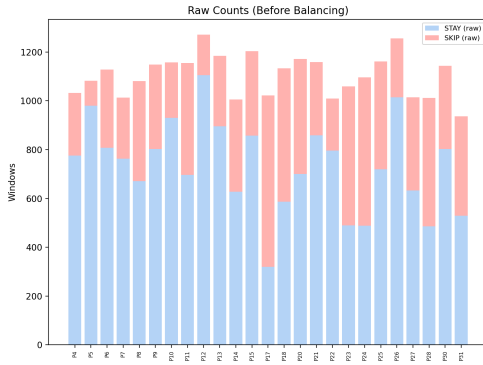


Figure 3.16: Raw window counts before balancing showing the extreme STAY dominance.

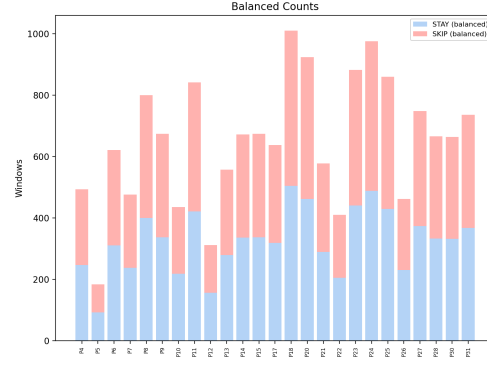


Figure 3.17: Window counts after undersampling, achieving a strict 50/50 ratio per participant.

Two cross-validation schemes are used throughout. The intra-subject setting applies blocked temporal 5-fold cross-validation within each participant; the inter-subject setting applies Leave-One-Group-Out cross-validation across participants, training on 24 participants and evaluating on the held-out 25th. To mitigate temporal leakage due to autocorrelation between adjacent EEG windows in the intra-subject setting, blocked temporal 5-fold cross-validation was applied. Each participant’s data were partitioned chronologically into five folds with equal counts of minority-class (**SKIP**) instances. Figure 3.18 shows minority-class occurrences (red) and fold boundaries (black), defined by equal splits of these events, per participant.

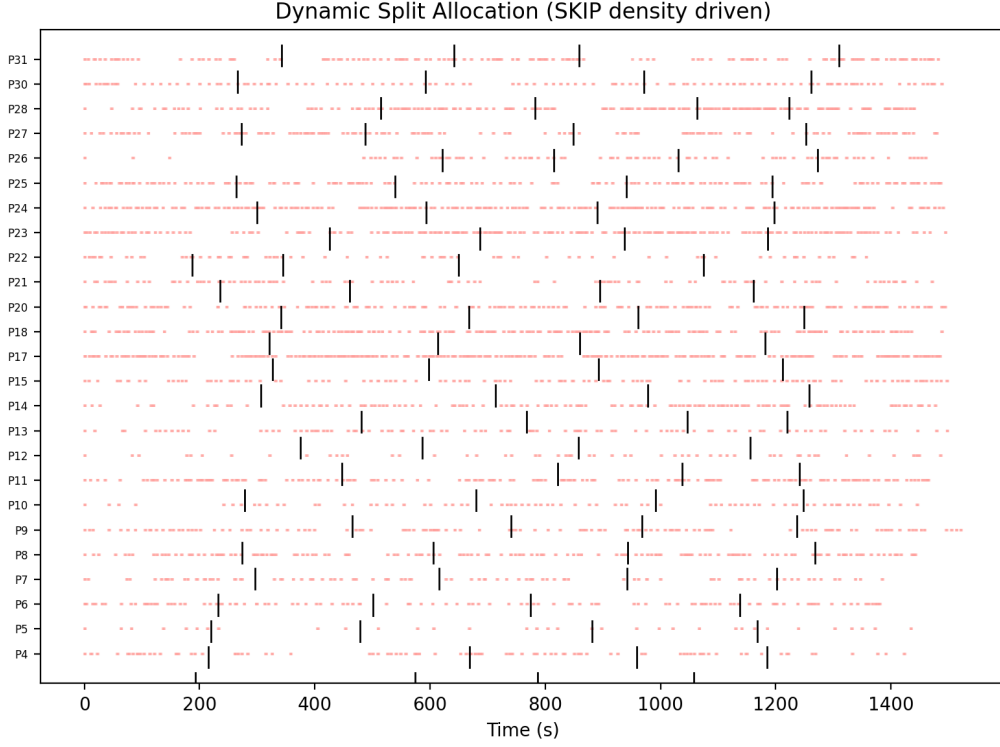


Figure 3.18: Each participant’s data is temporally split into 5 folds containing equal amounts of the minority SKIP class.

To rigorously prevent temporal leakage, a minimum separation of 3 seconds between all training and testing windows was strictly enforced at every fold boundary. This constraint was implemented as a hard criterion: any violation automatically triggered pipeline termination, ensuring that no autocorrelated samples could enter the training or test sets. As verified in Figure 3.19, all fold boundary gaps (red points along the x-axis) exceed the predefined 3-second threshold (red dashed line), confirming compliance across the dataset. This procedure produces valid minority-balanced folds while preserving strict independence of the test data; an example for a single participant is shown in Figure 3.20, where the separation gaps between folds (colored blocks) are clearly visible.

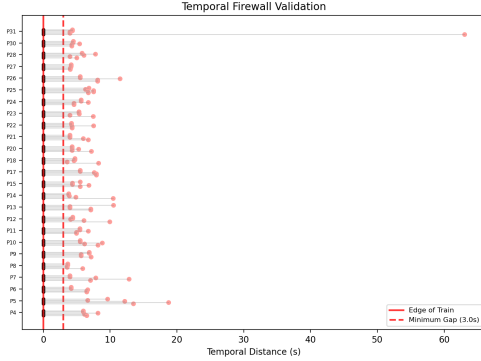


Figure 3.19: Temporal minimum required safety distance of 3 seconds between train and test windows is strictly fulfilled.

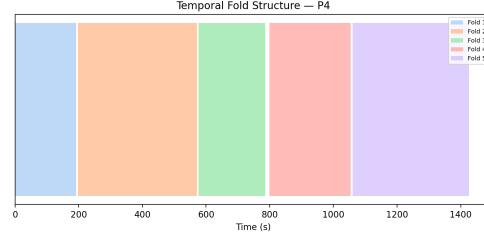


Figure 3.20: Temporal cross-validation split visualization, detailing the visible gaps between folds that successfully prevent data leakage.

3.8.2 Feature Separation Analysis

To mathematically evaluate the discriminative power of individual features prior to modeling, Cohen’s d was computed to measure the standardized effect size between the **STAY** and **SKIP** distributions for each engineered feature (executed via https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_analysis/viz17_feature_ranking.py). Cohen’s d is computed as defined in Equation 3.1. As a standardized effect size, Cohen’s d provides a scale-invariant measure of class separation that is independent of the absolute units of the feature (e.g., microvolts versus power spectral density). It quantifies how many standard deviations separate the centers of the two classes, thereby capturing theoretical separability prior to any model training.

By calculating the absolute effect size $|d|$, the pipeline systematically ranked all extracted features (e.g., TP10_raw_std) according to their individual ability to separate the **STAY** and **SKIP** classes. Aggregated plots grouped by frequency band, electrode, or statistic report the arithmetic mean of $|d|$ across all constituent features in that category and are shown without confidence intervals; the band $\times$ electrode profile is shown as a point plot with bootstrapped 95% confidence intervals.

Confidence intervals are reported only for this band-electrode profile, where each point represents a single well-defined feature combination; category-aggregated means (by band, electrode, or statistic alone) intentionally omit confidence intervals, as they average over heterogeneous and unequally-sized feature sets whose spread reflects between-feature heterogeneity rather than

sampling uncertainty of a single estimand. Ranked individual-feature ($|d|$) values (Table 4.1) are likewise reported as a descriptive screening criterion for candidate selection; formal inferential claims of superiority over baseline models are instead established via the Wilcoxon signed-rank tests in Section 3.8.4.

Feature-level rankings report the exact $|d|$ of a single named feature. Because the raw broadband signal contributes three statistics per electrode whereas each frequency band contributes eight, category means rest on unequally sized feature sets, and comparisons between them are indicative rather than exact. Features demonstrating the highest absolute $|d|$ values were designated as Top Features and isolated for independent predictive baseline modeling.

3.8.3 Logistic Regression Modeling and Baselines

To benchmark the true predictive capability of the engineered EEG features against chance and clinical baselines, three distinct modeling approaches were evaluated on the temporally isolated test sets (executed via https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_analysis/step07_train_intra.py and synthesized in https://github.com/coffeeandcoffee/eeg_tiktok_pipeline/data_analysis/step16_parallel_universes.py).

1. Coinflip Baseline: A uniform dummy classifier (`DummyClassifier(strategy='uniform')`) was utilized to establish an empirical chance baseline. Given the rigorous 50/50 balancing of the datasets, this approach reliably generated predictions hovering around the 50% chance level, verifying that no underlying structural bias existed within the cross-validation folds.

2. Engagement Index (EI) Baseline: The pre-calculated EI scalar (derived from $\beta/(\alpha + \theta)$) was scaled utilizing a standard scaler (subtracting the mean and scaling to unit variance) and fed as a singular feature into a Logistic Regression model (`LogisticRegression(class_weight='balanced')`). This established a clinical baseline, indicating how well classical engagement theory alone could predict the behavioral swipe decision in this specific short-form video context.

3. Top Feature Logistic Regression: To test the predictive power of individual statistical features independently, the features ranked highest via Cohen’s d (e.g., specific band powers or statistical moments) were similarly scaled and evaluated using isolated Logistic Regression models. This isolated approach allows for direct comparison of individual feature efficacy without

the confounding complexities of high-dimensional multivariate models.

To ensure a strict one-to-one comparison with the one-dimensional Engagement Index baseline, each macro “Top Parameter” (band, electrode, statistic) was represented by its strongest single constituent feature.

3.8.4 Metrics and Significance Testing

To comprehensively evaluate model performance, standard classification metrics were computed for each cross-validation fold: Accuracy, Precision, Recall, and the F_1 -score. In this context, correct prediction of the **STAY** class reflects successful detection of continuous non-skip viewing, whereas correct prediction of the **SKIP** class corresponds to identifying the period preceding a swipe-away decision within a 2–5 second pre-action window.

The F_1 -score was used as the primary performance metric, as it balances Precision and Recall and is robust to class imbalance. Metrics were computed separately for each class by treating the respective class as the positive label (i.e., one-vs-rest evaluation for **SKIP** and **STAY**).

Metric definitions: Let TP, FP, TN, and FN denote true positives, false positives, true negatives, and false negatives, respectively. To achieve optimal classification performance, the correct predictions (TP and TN) must be maximized while the incorrect predictions (FP and FN) are minimized.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Case 1: **SKIP** as the positive class

- **TP:** correctly predicted **SKIP** instances
- **TN:** correctly predicted **STAY** instances
- **FP:** **STAY** instances incorrectly predicted as **SKIP**
- **FN:** **SKIP** instances incorrectly predicted as **STAY**

Case 2: STAY as the positive class

- **TP:** correctly predicted STAY instances
- **TN:** correctly predicted SKIP instances
- **FP:** SKIP instances incorrectly predicted as STAY
- **FN:** STAY instances incorrectly predicted as SKIP

A class-specific F_1 -score substantially exceeding chance level indicates that the model captures discriminative structure beyond random guessing for that class. For reference, under balanced random predictions, the expected F_1 -score is approximately 0.5, whereas under strong class imbalance, this baseline decreases and should be interpreted relative to the class distribution.

To rigorously evaluate whether the Logistic Regression models utilizing individual EEG features outperformed the established baselines across the participant cohort, non-parametric statistical testing was employed. Specifically, the Wilcoxon signed-rank test was used to compute the statistical significance (p -value) of the performance differences between paired model distributions (e.g., Top Feature model vs. EI Baseline model, or EI Baseline model vs. 50% chance level) at the participant level. For each participant, the performance metric (e.g., F_1 -score) of the evaluated model was contrasted with that of the baseline, yielding 25 paired differences, one per participant.

The algorithm employs a two-sided Wilcoxon test with a standard significance threshold ($\alpha = 0.05$). Because a two-sided test detects significant differences in either direction, a directional validation check is applied post hoc: a model is deemed statistically significantly superior only if two conditions are met simultaneously, namely (i) the test yields $p < 0.05$, and (ii) the arithmetic mean of the model’s scores across all participants is strictly greater than the arithmetic mean of the corresponding baseline. The rank-biserial correlation (r) is also computed to quantify the effect size of the Wilcoxon test, indicating the strength of the model’s superiority over the compared baseline. This structured chain of statistical proof robustly determines whether individual EEG features can provide a significant improvement over both pure chance and classical clinical engagement metrics when attempting to predict real-world short-form video interactions. Because these comparisons are repeated across a number of feature–baseline pairs, metrics, and evaluation settings, the reported p -values are not corrected for multiple comparisons; individual comparisons are therefore best read as exploratory

screening criteria rather than confirmatory hypothesis tests, with family-wise error control identified as a priority for future replication (see Future Research).

Chapter 4

Results and Discussion

This chapter presents the results of a multi-stage experimental investigation. This thesis delivers five core scientific contributions:

1. **Primary EEG Experiment under Naturalistic Conditions**

The foundation of this work is the primary experiment (Methods Section 3.1 - 3.3, Results Section 4.1), in which participants were recorded via EEG while engaging in naturalistic TikTok browsing. This data collection phase constituted a substantial portion of the overall effort, requiring careful experimental design, technical implementation, and data acquisition under ecologically valid conditions. Not only the resulting dataset, but also the easily reproducible nature of the developed experimental method forms the basis for all subsequent analyses and should be understood as a central contribution of this thesis.

2. **Reusable EEG Processing and Analysis Pipeline**

In addition to the experimental dataset and process, a further key contribution lies in the development of a comprehensive script library and analysis pipeline (Methods Sections 3.4–3.8, Results Section 4.2). This framework enables fully reproducible processing, from automated EEG recording on standard laptop hardware to feature extraction, class separation, and machine learning-based prediction. As such, it not only supports the present analysis but also provides a transferable methodological toolset that can be readily adapted to related experimental paradigms.

3. **Behavioral Swipe-Burst Phenomenon**

We continue by focusing on the behavioral burst analysis (Methods Section 3.6.5, Results Section 4.3). As no burst rejection was implemented—due to scope constraints outlined in Section 3.6.5—this component is

treated as a secondary result rather than the central analytical pathway.

4. Evaluating the Engagement Index’s Predictive Capacity

The primary results emerge from a multi-layer analysis (Methods Sections 3.7–3.8, Results Section 4.4): Using the Cohen’s d method described in Section 3.8.2, first class separation capability of the conventional Engagement Index was evaluated, then specific features were identified that most effectively differentiate between the SKIP and STAY classes (Section 4.4.1) and which features outperform the Engagement Index’s class-separation capability of SKIP and STAY. Building on this, a set of machine learning models as described in Section 3.8.3 is applied to predict scrolling behavior based on these class distinctions using the Engagement Index with a Logistic Regression and comparing it to a random-chance baseline, as well as challenging the conventional Engagement-Index by using previously identified superior features with the same Logistic Regression approach and comparing it to the Engagement Index. Best-performing models and features are reported (Section 4.4.2).

As described in Section 3.8.4, these results are carefully introduced along their statistical significance, a crucial measure considering the sample size of $n=25$ participants, in order to give result metrics such as the F1-score the appropriate context for meaningful interpretation for discussion and further use. While this prediction task serves as an initial proof of concept, it already indicates which features may provide sufficient discriminatory power.

5. Neuroscientific and Behavioral Interpretation

The results (Section 4.4.1) offer insights into the brain regions and frequency bands most predictive of this behavior, thereby informing both theoretical interpretation and potential practical applications. Therefore, in the final section (Section 4.5), these findings are interpreted in the context of neuroscience, psychology, and behavioral research, highlighting their implications and real-world relevance.

4.1 Key Findings from the Experimental Paradigm

The experimental paradigm proved feasible in a semi-controlled but still naturalistic setting, and several practical lessons emerged from its execution.

Recruitment via email outreach and student mailing lists worked well, especially when paired with a clear and modest incentive. Participants were generally receptive to the study, and most were academically oriented, which appeared to increase interest in the research question and reduce concerns about signing the consent and data-protection forms.

A particularly useful element was the use of iOS Guided Access during both the baseline and TikTok phases. It provided a simple way to constrain the task environment, prevent app switching, and automatically track the duration spent in each app, which matched the temporal structure of the design well. This was especially helpful for the 2-minute baseline and the 25-minute scrolling phase, where reliable timing was essential.

At the same time, the study also highlighted several practical challenges. One recurring difficulty was maintaining a strict and consistent script during participant instruction, especially when explaining what participants should do when the task started, paused, or ended. Another challenge was ensuring that the instructions around synchronization, keypress behavior, and task boundaries were communicated clearly and uniformly across sessions. These issues did not invalidate the experiment, but they do suggest that the protocol would benefit from a more standardized briefing script and a more formalized session flow in future replications.

4.2 The Reusable Methodological Pipeline

A major practical contribution of this thesis is the reusable analysis and recording pipeline, which can be accessed through the associated GitHub repository, first mentioned in Section 3.2 and further described in Section 3.5. In the present study, the pipeline ran mostly smoothly after multiple months of optimization, and finally enabled the full workflow from recording to preprocessing, feature extraction, and analysis. The main technical interruptions were occasional Bluetooth dropouts, which were handled by the recording architecture without causing major session failures, but which still underline the need for continued robustness improvements.

The development process also revealed an important software-engineering lesson: a substantial portion of the code base became obsolete during the iterative refinement across multiple months. Precisely 69.6% (representing a reduction from 37,189 to 11,311 total lines of code) of the scripts were temporary, diagnostic, or superseded by improved versions during development. While this is normal in an exploratory research pipeline, it also shows that the project would benefit from a more systematic cleanup strategy in future iterations. A lightweight script-archiving or pruning mechanism should

therefore be introduced next time to preserve clarity, reduce maintenance burden, and make the final codebase easier to reuse and extend.

More broadly, the pipeline demonstrates that a relatively compact and open methodological framework can support consumer-grade EEG recording, synchronized behavioral logging, preprocessing, and downstream feature analysis in a reproducible way. That makes it not only useful for this thesis, but also transferable to other studies with similar naturalistic EEG and behavior-synchronization requirements.

4.3 Behavioral Swipe-Burst Phenomenon

As established in the methodology (Section 3.6.5), a burst is defined as a sequence of rapidly successive skip events reflecting frantic swiping. To quantify this burst severity, overlapping SKIP windows were grouped into “burst blocks.” The block size n denotes the number of consecutive rapid swipes before a participant pauses to engage. A block of $n = 1$ represents standard behavior, whereas larger values (e.g., $n = 3$) indicate repeated rapid skipping, possibly without meaningful engagement due to high frequency and action-state inertia.

Figure 4.1 shows this behavior at the participant level, revealing a clear dichotomy: some participants exhibit frequent large burst blocks (“fast scrollers”, e.g. P17), while others predominantly show $n = 1$ behavior (“careful scrollers”, e.g. P15). Figure 4.2 aggregates these patterns, confirming that while $n = 1$ dominates, a long tail of burst behavior persists.

Finally, Figure 4.3 illustrates the sensitivity to the chosen window size by repeating the analysis with a ± 5.0 -second window (10.0-second merge threshold). The wider threshold captures a larger portion of the dwell time distribution and merges many isolated swipes into larger burst blocks, substantially reducing the proportion of $n = 1$ events, simply by changing the definition of a “burst block”. Specifically, under the default ± 3.0 s window, 67.4% of swipe events are isolated singletons (participant-level mean $69.1\% \pm 16.2\%$), while widening to ± 5.0 s reduces this to 48.9% (participant-level mean $46.9\% \pm 20.8\%$). The large between-participant standard deviation is itself informative: burst tendency appears to be a strongly individual trait rather than a fixed population parameter, foreshadowing the precision-public-health argument developed in Section 4.5. Specifically, under the ± 3.0 s window paradigm, 67.4% of all swipe events are isolated $n = 1$ sequences (participant-level average: $69.1\% \pm 16.2\%$). In contrast, widening the window to ± 5.0 s reduces the overall proportion of isolated swipes to 48.9% (participant-level average: $46.9\% \pm 20.8\%$). This underscores the strong dependence on

parameter selection and the need for systematic optimization in future work.

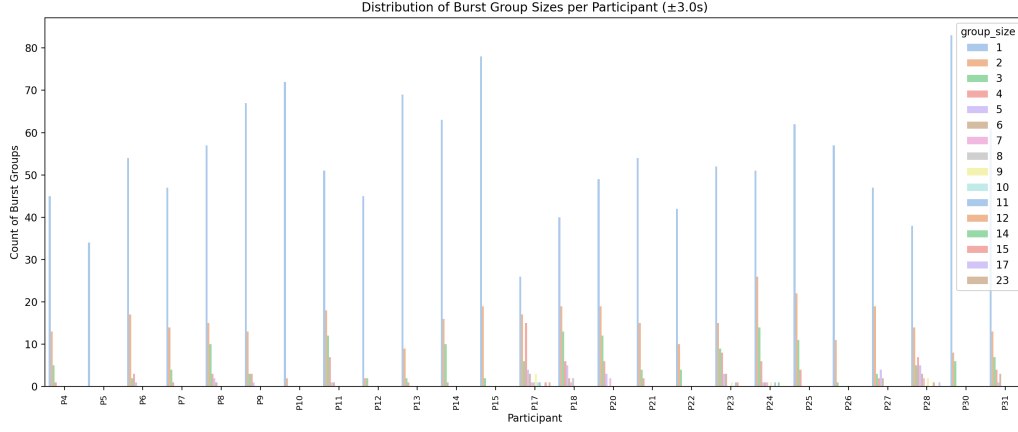


Figure 4.1: Distribution of burst group sizes per participant (using the $\pm 3.0s$ window). This highlights the distinct behavioral difference between “frantic scrollers” (larger burst blocks) and “careful scrollers” (predominantly isolated $n = 1$ swipes).

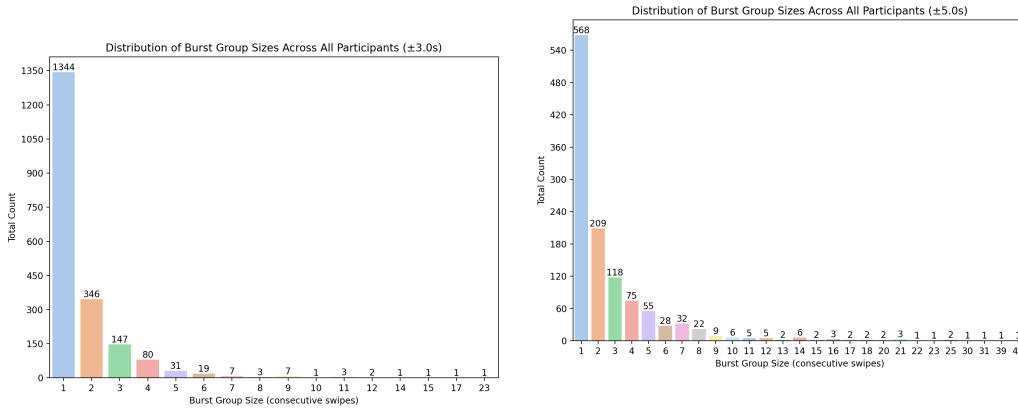


Figure 4.2: Total burst group sizes across all participants ($\pm 3.0s$ window).

Figure 4.3: Total burst group sizes across all participants ($\pm 5.0s$ window). The wider threshold computationally merges many normal swipes into larger burst blocks.

4.4 The Engagement Index’s Swipe-Prediction Performance

4.4.1 Feature Importance and Neurophysiological Correlates

In Figure 4.4, we observe that class separation, quantified via Cohen’s d as described in Section 3.8.2, is consistently stronger for several individual frequency bands (blue bars) than for the conventional Engagement Index (red dotted line). In particular, the α -band (8–13 Hz) achieves an effect size approximately 0.005 standard deviations higher than the Engagement Index, and the high- γ band (40–60 Hz), as the strongest candidate, exceeds it by more than 0.03 standard deviations. The Engagement Index — which combines three bands into a single scalar — itself reaches just above 0.02 and is therefore weaker in class separability than any of the individual frequency bands and the raw signal. This suggests that aggregating multiple bands into a single composite metric may dilute discriminative information that is more clearly expressed within specific, well-chosen frequency ranges.

Moreover, class separation reaches values above 0.04 only for a small subset of bands: the raw, broadband signal (without frequency decomposition), the δ band (1–4 Hz), the low- γ band (30–40 Hz), and the high- γ band (40–60 Hz). These results indicate that both very low-frequency activity and specific γ -range oscillations carry comparatively strong discriminative structure for the SKIP versus STAY decision. This is consistent with evidence that delta oscillatory responses are involved in decision-making and attentional processes during cognitive tasks [26], and that delta activity is implicated in salience detection, becoming functionally more prominent when higher-frequency, more operationally advanced processing is less dominant [27] — a pattern structurally consistent with the present finding that band-specific features outperform the composite, higher-frequency-weighted Engagement Index.

Taken together, these findings highlight three key points. First, individual frequency bands exhibit stronger class-separation capability than the conventional Engagement Index, suggesting that analyses based on raw or band-specific signals may be more informative than combined engagement metrics. Second, the raw broadband signal reaches a higher mean $|d|$ than all separated bands except high γ , consistent with prior EEG decoding work showing broadband/undecomposed signal can match or exceed band-limited features depending on task [33]. This comparison should be read with care, since the raw signal is summarized by only three time-domain statistics whereas each

band is summarized by eight (Section 3.7); it nevertheless motivates closer examination of both broadband information and high- γ activity in particular.

P8 Third, the pattern across bands follows a clear up-down-up trend: separability is high at low frequencies, drops to a minimum in the α range (8–13 Hz), then rises to a global maximum in the high- γ range before decreasing again for very high γ frequencies (60–100 Hz). Non-monotonic, band-specific separability profiles of this kind are not without precedent in EEG classification more broadly; band-wise decoding performance has been shown to peak in non-adjacent frequency ranges rather than scaling smoothly with frequency [35]. This systematic profile offers a frequency-resolved view of where discriminative information is concentrated in the EEG.

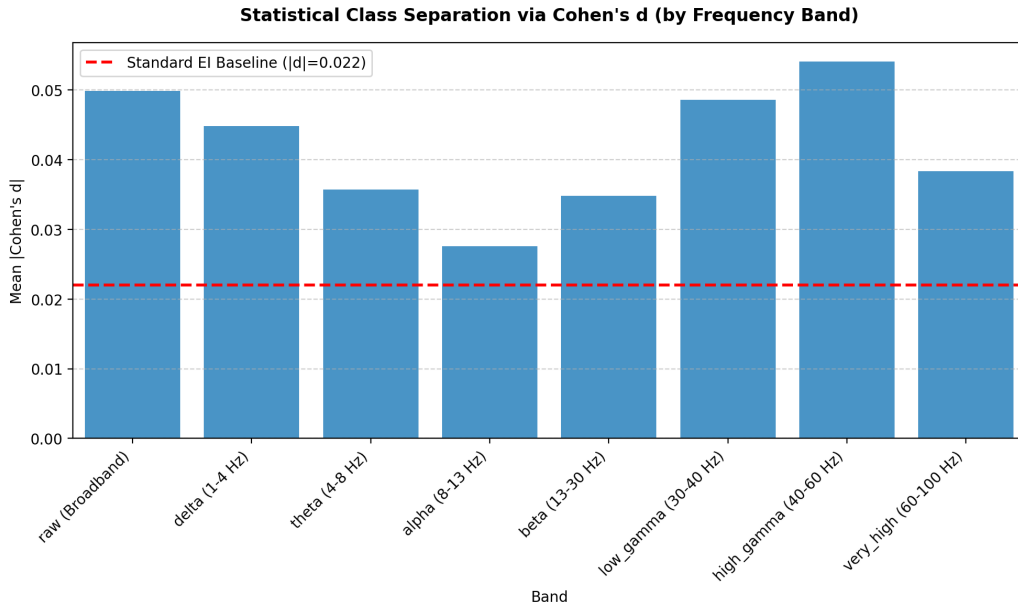


Figure 4.4: Mean $|d|$ by frequency band, aggregated across electrodes and statistics. The dotted line marks the Engagement Index reference value; error bars are omitted.

Similarly, Figure 4.5 shows that separating the signal by electrode of origin — TP9, AF7, AF8, and TP10 — yields higher class separability than the conventional Engagement Index in all cases. The improvement ranges from just under 0.01 standard deviations for TP9 and AF8 to more than 0.03 standard deviations for AF7 and TP10. This suggests that distinguishing between individual brain regions provides more informative class separation than collapsing the signal into a single composite metric, and that the dif-

ferent electrodes contribute unequally to this discriminative task.

Moreover, the figure reveals a clear dominance of AF7 (left anterior frontal) and TP10 (right temporal pole) over TP9 (left temporal pole) and AF8 (right anterior frontal) in terms of Cohen’s d . This pattern suggests that the frontal and right temporal regions contribute more strongly to the SKIP-versus-STAY separation than the left temporal pole and right anterior frontal regions, pointing to a potentially lateralized and region-specific neurophysiological signature of scrolling behavior.

Among the electrodes, AF7 (left anterior frontal) reaches the highest mean $|d|$, exceeding the value shown for the raw signal in Figure 4.4 by just over 0.005 standard deviations. This comparison is indicative rather than exact, because the two means aggregate over different and unequally sized feature sets: the electrode mean averages across all signal types and statistics at AF7, whereas the raw-signal mean averages across all electrodes but over only the three time-domain statistics extracted for the broadband signal (Section 3.7).

Potential systematic differences related to the recording hardware cannot be excluded at this stage, as two physical units of the headband were used across the $n = 25$ participants (Section 3.2).

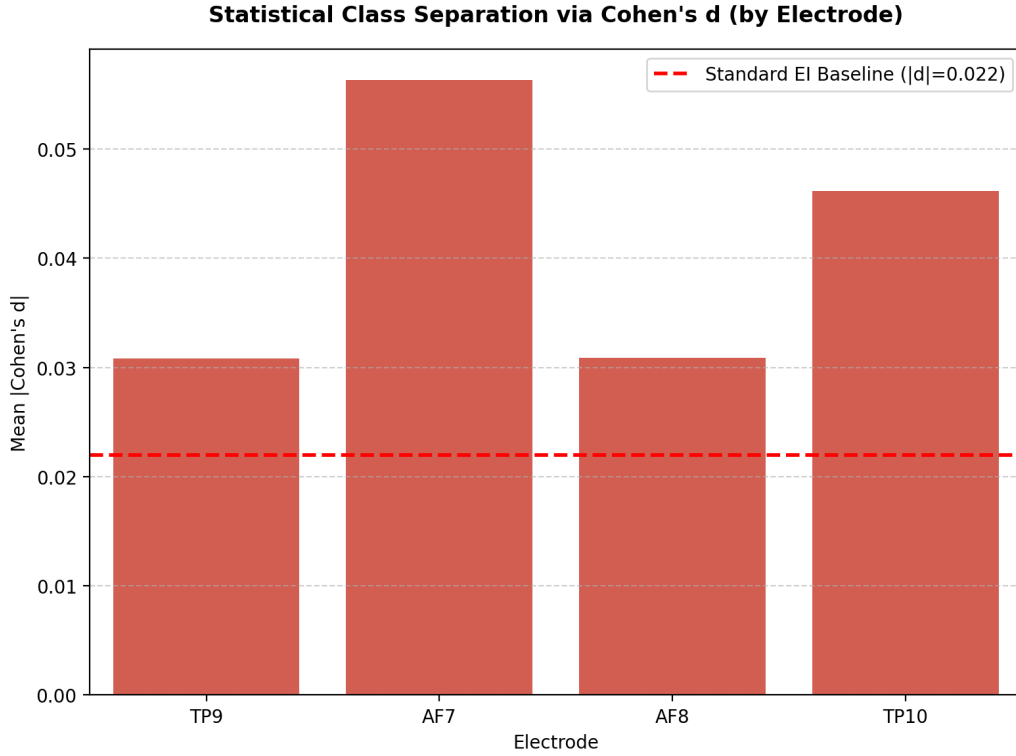


Figure 4.5: Mean $|d|$ by electrode, aggregated across signal types and statistics. The dotted line marks the Engagement Index reference value; error bars are omitted.

Figure 4.6 shows that, among the statistical moments and related features considered, only a small subset exceeds the conventional Engagement Index by at least 0.01 standard deviations in terms of Cohen’s d class separation. These features are Hjorth activity, Hjorth mobility, the mean, the standard deviation, Hjorth complexity, and peak frequency. Taken together, this suggests that simple time-domain descriptors and Hjorth-derived measures retain meaningful discriminative information, but only for a limited number of feature types.

Furthermore, only two statistics - Hjorth complexity and peak frequency - reach a higher mean $|d|$ than the raw signal, again noting that these means aggregate over different feature sets and are therefore only indicatively comparable. This indicates that, while the unprocessed signal already contains substantial discriminative structure, certain higher-order descriptors can still extract additional information beyond the broadband signal alone.

Across the three feature families examined here—frequency bands, elec-

trodes, and statistical moments/metrics—the statistical moments and metrics show the strongest overall class-separation performance. In particular, peak frequency stands out with a mean $|d|$ of nearly 0.07, placing it almost 0.05 standard deviations above the conventional Engagement Index and nearly 0.02 above the mean for the raw signal. This makes peak frequency the strongest single feature in the present comparison and suggests that oscillatory structure within the signal may be especially informative for distinguishing between the two classes. At the same time, these findings should be interpreted cautiously, as the observed advantages are modest in absolute terms and may be influenced by dataset-specific factors.

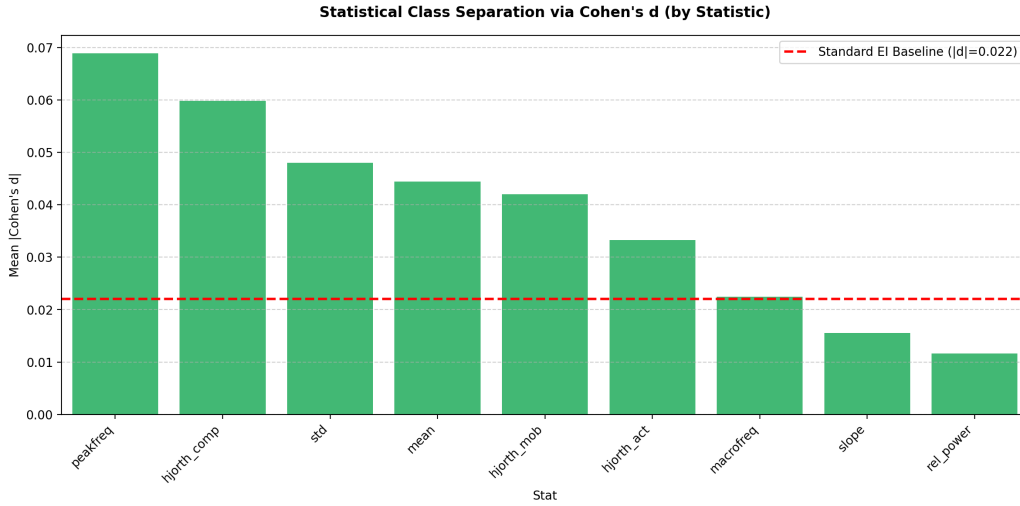


Figure 4.6: Mean $|d|$ by statistic, aggregated across electrodes and signal types. The dotted line marks the Engagement Index reference value; error bars are omitted.

While the comparison of features up to this point has already provided informative detail, the highest level of resolution is reached when each parameter is specified individually: (i) electrode, (ii) band or raw signal, and (iii) statistic or metric. Table 4.1 illustrates the value of this finer-grained perspective. In the preceding analyses, peak frequency emerged as the strongest class separator according to Cohen’s d , when considering all bands and electrodes jointly. In those comparisons, the only variable under inspection was the parameter being held constant, while the remaining parameters varied across features. This resulted in aggregated Cohen’s d values for each parameter group, as shown in Figures 4.4, 4.5, 4.7, and 4.6. These results raise a natural question: are there individual features, defined across all three

parameters, that exceed the currently best observed class-separation performance of peak frequency, at approximately 0.07?

Table 4.1 shows that such features do indeed exist. In fact, the top 30 features all outperform the peak-frequency result in terms of Cohen’s d , indicating that a more specific feature definition can yield substantially stronger class separability. The strongest feature is TP10 raw standard deviation, with a Cohen’s d of 0.196, corresponding to an improvement of nearly ten-fold over the conventional Engagement Index in terms of class separation.

This shows that class separability is not only determined by broad feature families such as frequency bands, electrodes, or statistical moments, but can increase substantially when the signal is described at the level of individual feature combinations. It further suggests that the conventional Engagement Index captures only part of the discriminative structure present in the data, whereas more specific features can reveal considerably stronger separation between classes. At the same time, these findings should be interpreted as evidence of dataset-specific predictive potential rather than as a universal hierarchy of feature importance.

Rank	Feature Name	Cohen's d
1	TP10 raw std	0.196
2	AF7 high gamma mean	0.168
3	AF7 high gamma hjorth mob	0.145
4	AF7 delta hjorth comp	0.141
5	AF8 delta peakfreq	0.140
6	AF7 high gamma peakfreq	0.139
7	TP9 delta hjorth comp	0.137
8	TP9 delta peakfreq	0.136
9	TP10 delta hjorth comp	0.131
10	TP10 delta peakfreq	0.130
11	AF7 theta hjorth comp	0.129
12	AF7 raw std	0.129
13	AF7 theta peakfreq	0.124
14	AF7 low gamma hjorth comp	0.124
15	AF7 delta peakfreq	0.123
16	TP10 low gamma hjorth act	0.119
17	AF8 high gamma mean	0.119
18	AF7 low gamma mean	0.118
19	AF7 beta mean	0.115
20	AF8 delta hjorth comp	0.115
21	TP10 low gamma std	0.114
22	TP10 high gamma hjorth mob	0.110
23	AF7 very high mean	0.109
24	TP10 beta hjorth act	0.108
25	TP9 theta peakfreq	0.103
26	TP9 high gamma std	0.103
27	TP10 beta std	0.100
28	AF8 theta peakfreq	0.100
29	AF7 beta hjorth comp	0.100
30	TP9 high gamma hjorth act	0.098

Table 4.1: Top 30 Features by Separation (|Cohen's d |).

For all features, Figure 4.7 shows Cohen's d values sorted jointly by frequency band and electrode. For the raw, broadband signal (no frequency filtering), the TP10 combination shows the highest mean $|d|$ in the visualization, consistent with the top individual feature "TP10 raw std" originating from this combination. Overall, similar trends to the previous figures can be observed, but the combined view makes the joint band-electrode profile more apparent. In particular, there is a pronounced spike in class-separation

performance for the broadband signal at TP10 and AF7.

Moreover, the AF7 electrode stands out by exhibiting a clear upward trend in Cohen’s d from the α band (8–13 Hz) through β (13–30 Hz) to low γ (30–40 Hz), followed by a plateau across the broader γ range (30–100 Hz; low, high, and very high γ) with a slight decline toward the highest frequencies. The peak for AF7 lies in the low- γ band (30–40 Hz), where its class-separation ability is highest. This enhanced separability for the SKIP-versus-STAY distinction at AF7 (left anterior frontal), particularly in the γ and high- γ range (30–100 Hz), may suggest that left anterior frontal oscillatory activity in this frequency range is especially sensitive to the neural processes that differentiate **STAY** from **SKIP** behavior in this task. These interpretations should, however, be viewed as preliminary and require replication in independent datasets.

Furthermore, the superior class-separation ability of the TP10 broadband signal, relative to TP10’s performance within individual frequency bands, suggests that discriminative information may extend beyond the conventional 1–100 Hz decomposition. This could indicate that additional signal components, including potential artifacts such as electromyographic activity from jaw movements, contribute to the SKIP-versus-STAY separation alongside, or in some cases instead of, purely neural activity. This possibility is not merely speculative: temporal electrode sites are known to be disproportionately susceptible to broadband EMG contamination from the underlying temporalis muscle, an effect that is present even at rest, spans the same broadband/high-frequency range implicated here, and survives standard ICA-based correction [23–25]. These possibilities highlight the need for careful artifact characterization and control in future work.

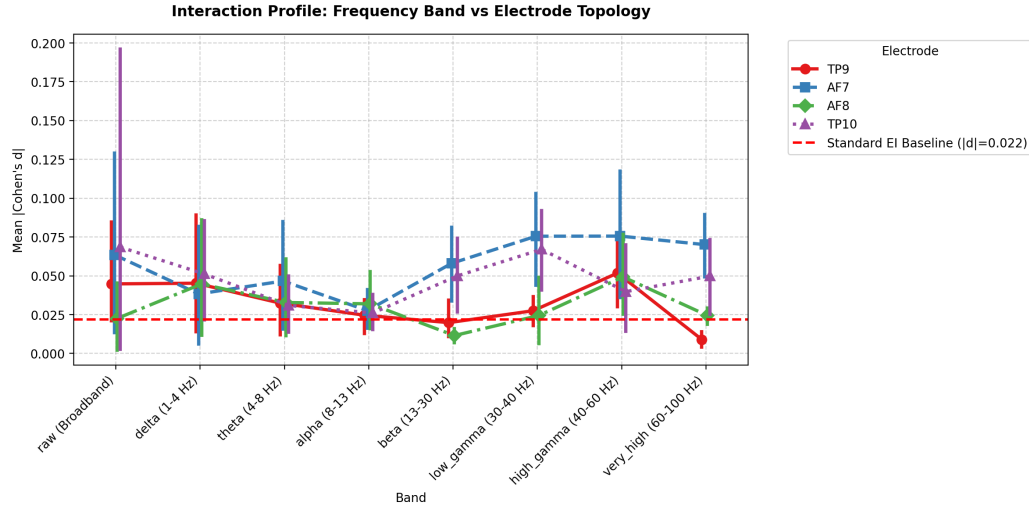


Figure 4.7: Mean $|d|$ for each frequency-band $\times$ electrode combination (band–electrode separation profile). Error bars represent bootstrapped 95% confidence intervals.

As shown in Figure 4.8 the top features were selected for each category: i) overall top feature #1, ii) overall top feature #2, iii) top #1 feature of top band, iv) top #1 feature of top electrode, v) top #1 feature of top statistic/metric.

The feature “TP10 raw std” ($|d| = 0.196$) is placed for i) feature #1, whereas “AF7 high gamma mean” ($|d| = 0.17$) simultaneously represents all 3 categories ii), iii) and iv). The “AF8 delta peakfreq” ($|d| = 0.14$) feature takes the place of category v), which is the #1 feature amongst the top statistics, namely the Peak-frequency.

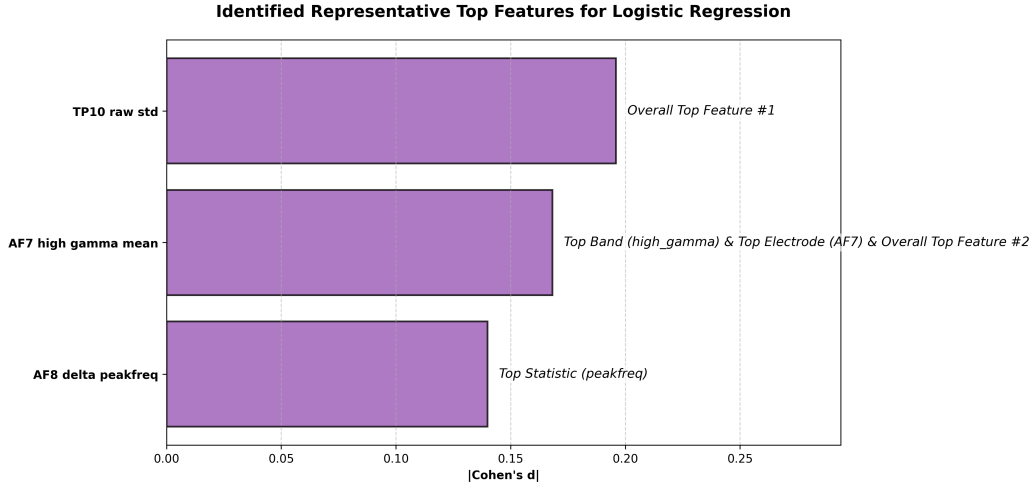


Figure 4.8: Selection of the three candidate features carried forward to the Alternative Index models: overall top feature, top feature of the strongest band, and top feature of the strongest statistic.

4.4.2 Engagement Index Model Performance and Benchmark Comparison

The modeling framework described in Section 3.8.3 was used to evaluate whether the engineered EEG features could predict **SKIP** versus **STAY** behavior beyond chance and beyond the conventional Engagement Index. First, a coin-flip baseline established the empirical chance level. Second, the Engagement Index was evaluated as a one-dimensional logistic regression baseline. Third, the strongest features identified in Section 4.4.1 were tested in isolated logistic regression models to assess whether they could improve predictive performance beyond the baseline models.

To evaluate predictive performance, standard classification metrics were computed as described in Section 3.8.4. In particular, Accuracy, Precision, Recall, and the F_1 -score were calculated for each cross-validation fold, separately for both classes via one-vs-rest evaluation. The F_1 -score served as the primary metric because it balances Precision and Recall and is therefore more informative than Accuracy under the present class structure.

Statistical significance was assessed using the Wilcoxon signed-rank test, as outlined in Section 3.8.4. The coin-flip baseline serves as the reference for chance-level performance, whereas the Engagement Index is used as the clinical baseline for subsequent comparisons. Performance is reported both for the intra-subject evaluation setting, in which models were trained and tested

within the same participant, and for the inter-subject evaluation setting, in which models were generalized across participants. The corresponding results are shown in Table 4.2 and Table 4.3, respectively. In these tables, green values indicate metrics that are both significantly different from the baseline ($p < 0.05$) and higher than the baseline on average, thus reflecting genuine performance gains rather than mere statistical deviation.

At the inter-subject level shown in Table 4.3, the Engagement Index logistic regression yields a statistically significant improvement over the coin-flip baseline only for **STAY** as the positive class. The most prominent effect is Recall, which rises to 89.2% compared with 51.7% for the coin-flip baseline. This indicates that the model is highly sensitive to **STAY** windows and rarely misses them, but it does not necessarily imply balanced or reliable class discrimination on its own. In fact, high Recall in combination with only moderate F_1 -scores suggests a strong tendency to capture the majority of **STAY** instances while still leaving room for false positives, meaning that the model may be better at recognizing **STAY** periods than at sharply distinguishing them from **SKIP**.

The corresponding F_1 -scores provide a more nuanced interpretation. At 62.0% for training and 62.5% for testing, the model performs above chance and remains relatively stable across folds, which argues against severe overfitting in the inter-subject setting. However, the gain over the coin-flip baseline is modest in absolute terms, especially when viewed alongside the comparatively large improvement in Recall. This combination suggests that the Engagement Index carries some predictive structure, but that this structure is unevenly distributed across metrics: it may identify **STAY** behavior reasonably well, yet it does not provide a uniformly strong decision boundary between the two classes.

Importantly, these results also clarify what the Engagement Index can and cannot do in this dataset. It appears capable of capturing a limited but real signal related to the **STAY** class specifically, but it is not sufficient as a standalone descriptor of the full **SKIP**-versus-**STAY** decision. The fact that the model improves clearly in Recall while remaining only moderately strong in F_1 -score raises a central interpretive question: does the Engagement Index reflect a broad engagement-related tendency, or is it primarily sensitive to a subset of windows that are easier to classify as **STAY**? This ambiguity motivates the next analytical step in Section 4.4.3, namely to examine which specific features, electrodes, and frequency ranges contribute more directly to class separability and whether they offer a more interpretable explanation as an “Alternative Index” over the aggregate Engagement Index. In that sense, the present numbers do not merely indicate performance; they identify the precise limitation of the baseline and justify moving toward feature-level

interpretation and model comparison.

Metric	Coin Flip	EI LR	
	Val	Val	Sig
F1 Train (STAY)	50.6%	53.8%	TRUE
F1 Test (STAY)	52.4%	50.9%	TRUE
Precision (STAY)	50.1%	49.1%	TRUE
Recall (STAY)	55.0%	52.9%	TRUE
F1 Train (SKIP)	49.4%	49.6%	FALSE
F1 Test (SKIP)	47.5%	46.9%	FALSE
Precision (SKIP)	50.1%	48.9%	TRUE
Recall (SKIP)	45.2%	45.1%	FALSE

Table 4.2: Step 1 (Intra-subject): EI vs Coin Flip. Significance tested using Wilcoxon signed-rank test against Coin Flip (two-sided, $\alpha = 0.05$, N=25).

Metric	Coin Flip	EI LR	
	Val	Val	Sig
F1 Train (STAY)	50.3%	62.0%	TRUE
F1 Test (STAY)	50.8%	62.5%	TRUE
Precision (STAY)	49.9%	49.4%	FALSE
Recall (STAY)	51.7%	89.2%	TRUE
F1 Train (SKIP)	49.6%	20.6%	TRUE
F1 Test (SKIP)	48.9%	10.7%	TRUE
Precision (SKIP)	49.9%	23.5%	TRUE
Recall (SKIP)	48.0%	9.6%	TRUE

Table 4.3: Step 1 (Inter-subject): EI vs Coin Flip. Significance tested using Wilcoxon signed-rank test against Coin Flip (two-sided, $\alpha = 0.05$, $N=25$).

When assessing the individual participants’ performance in the cross-participant model setting, as shown in Figure 4.9, the pattern further supports the interpretation that **STAY** events are systematically and significantly easier for the Engagement Index to recognize than **SKIP** events are to detect reliably. This asymmetry suggests that the model captures a broad tendency toward the **STAY** class, but lacks equally strong sensitivity to the more transient **SKIP** class.

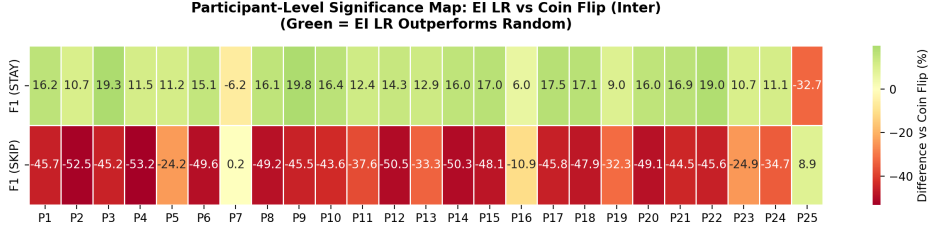


Figure 4.9: Inter-subject significance heatmap comparing F_1 -test performance of the Engagement Index and the coin-flip baseline for SKIP and STAY as positive classes.

4.4.3 Alternative Index Exploration

The Alternative Index exploration evaluates the three features with the highest $|d|$ identified in Section 4.4.1, namely “TP10 raw std”, “AF7 high gamma mean” and “AF8 delta peakfreq”, which are fed into a Logistic Regression model, and compared directly to the Engagement Index Logistic Regression model performance from the previous step in Section 4.4.2.

Intra-Subject (Per-Participant Level)

The performance of TP10 raw standard deviation relative to the Engagement Index is shown in Table 4.4. At the intra-subject level, the only significant improvement is observed for F_1 -train with SKIP as the positive class, where TP10 raw standard deviation reaches 53.6% compared with 49.6% for the Engagement Index. Because this advantage does not carry over to the test set, it should be interpreted cautiously and is more consistent with participant-specific overfitting than with robust predictive structure. Nevertheless, the result is still informative: it suggests that raw signal variability at TP10 may contain class-related information that the Engagement Index does not capture, even if that information is not stable enough to generalize within participants.

Metric	EI LR	TP10 raw std	
	Val	Val	Sig
F1 Train (STAY)	53.8%	52.4%	TRUE
F1 Test (STAY)	50.9%	49.3%	TRUE
Precision (STAY)	49.1%	48.2%	TRUE
Recall (STAY)	52.9%	50.4%	TRUE
F1 Train (SKIP)	49.6%	52.6%	TRUE
F1 Test (SKIP)	46.9%	46.8%	FALSE
Precision (SKIP)	48.9%	47.9%	TRUE
Recall (SKIP)	45.1%	45.7%	FALSE

Table 4.4: Step 5 (Intra): Proposed Feature (TP10 raw std) vs EI. Significance tested using Wilcoxon signed-rank test against EI (two-sided, $\alpha = 0.05$, $N=25$).

The performance of AF7 high- γ mean relative to the Engagement Index is shown in Table 4.5. Here, several small but significant improvements emerge: Precision for **STAY** rises to 49.9% from 49.1%, F_1 -train for **SKIP** increases to 52.9% from 49.6%, and Precision for **SKIP** reaches 49.8% compared with 48.9% for the Engagement Index. Although these gains are modest in absolute terms, their consistency across both classes suggests that AF7 high- γ activity may carry slightly more class-specific structure than the Engagement Index. In other words, this feature does not dramatically outperform the baseline, but it does appear to encode a more balanced and slightly more sensitive representation of the **SKIP**/**STAY** distinction.

Metric	EI LR	AF7 high gamma mean	
	Val	Val	Sig
F1 Train (STAY)	53.8%	52.6%	TRUE
F1 Test (STAY)	50.9%	52.1%	FALSE
Precision (STAY)	49.1%	49.9%	TRUE
Recall (STAY)	52.9%	54.6%	FALSE
F1 Train (SKIP)	49.6%	52.9%	TRUE
F1 Test (SKIP)	46.9%	47.2%	FALSE
Precision (SKIP)	48.9%	49.8%	TRUE
Recall (SKIP)	45.1%	45.0%	FALSE

Table 4.5: Step 5 (Intra): Proposed Feature (AF7 high gamma mean) vs EI. Significance tested using Wilcoxon signed-rank test against EI (two-sided, $\alpha = 0.05$, N=25).

The performance of AF8 delta peak frequency relative to the Engagement Index is shown in Table 4.6. In this case, both F_1 -train scores are significantly higher, reaching 83.0% and 83.1% for **SKIP** and **STAY** as the positive class, respectively. However, the corresponding test performance drops sharply toward chance level, which strongly suggests overfitting. Interestingly, F_1 -test, Precision, and Recall for the **SKIP** class are still slightly and significantly better than the Engagement Index, indicating that some task-relevant structure is present, even though it is not stable enough to generalize cleanly within participants. This pattern is important because it shows that a feature may carry real discriminative information while still being too context-specific to function as a dependable standalone predictor.

Metric	El LR	AF8 delta peakfreq	
	Val	Val	Sig
F1 Train (STAY)	53.8%	83.0%	TRUE
F1 Test (STAY)	50.9%	49.6%	TRUE
Precision (STAY)	49.1%	50.3%	TRUE
Recall (STAY)	52.9%	48.9%	TRUE
F1 Train (SKIP)	49.6%	83.1%	TRUE
F1 Test (SKIP)	46.9%	51.0%	TRUE
Precision (SKIP)	48.9%	50.3%	TRUE
Recall (SKIP)	45.1%	51.7%	TRUE

Table 4.6: Step 5 (Intra): Proposed Feature (AF8 delta peakfreq) vs EI. Significance tested using Wilcoxon signed-rank test against EI (two-sided, $\alpha = 0.05$, $N=25$).

Inter-Subject (Cross-Participant Level)

The performance of TP10 raw standard deviation relative to the Engagement Index is shown in Table 4.7. For **SKIP** as the positive class, all performance metrics improve significantly over the Engagement Index. F_1 -train rises from 20.6% to 52.4%, F_1 -test increases from 10.7% to 44.3%, Precision improves from 23.5% to 43.0%, and Recall rises from 9.6% to 50.8%. It is worth stating explicitly that an F_1 -test of 44.3% remains below the approximate 0.5 chance-level reference for balanced binary classification (Section 3.8.4); the gain reported here is therefore an improvement over a weak Engagement Index baseline rather than evidence of exceeding chance-level discrimination in the cross-participant setting. The pattern is nevertheless meaningful: unlike the Engagement Index, TP10 raw standard deviation appears substantially more sensitive to the transient **SKIP** state. This suggests that raw signal variability at TP10 may preserve information that is more directly aligned with the **SKIP** class than the aggregate Engagement Index score does.

Metric	EI LR	TP10 raw std	
	Val	Val	Sig
F1 Train (STAY)	62.0%	49.6%	TRUE
F1 Test (STAY)	62.5%	46.0%	TRUE
Precision (STAY)	49.4%	47.5%	FALSE
Recall (STAY)	89.2%	51.0%	TRUE
F1 Train (SKIP)	20.6%	52.4%	TRUE
F1 Test (SKIP)	10.7%	44.3%	TRUE
Precision (SKIP)	23.5%	43.0%	TRUE
Recall (SKIP)	9.6%	50.8%	TRUE

Table 4.7: Step 5 (Inter): Proposed Feature (TP10 raw std) vs EI. Significance tested using Wilcoxon signed-rank test against EI (two-sided, $\alpha = 0.05$, $N=25$).

The performance of AF7 high- γ mean relative to the Engagement Index is shown in Table 4.8. Again, the SKIP class is predicted more effectively than by the Engagement Index. F_1 -train increases from 20.6% to 45.5%, F_1 -test from 10.7% to 35.5%, Precision from 23.5% to 52.4%, and Recall from 9.6% to 42.7%. While this does not yet establish a fully reliable cross-participant classifier, it does show that AF7 high- γ activity captures a more informative decision-related pattern than the Engagement Index baseline. The improvement in Precision is particularly notable, as it suggests that when the model predicts SKIP, it does so with substantially greater confidence and reduced false-alarm behavior than the Engagement Index.

Metric	EI LR	AF7 high gamma mean	
	Val	Val	Sig
F1 Train (STAY)	62.0%	54.3%	TRUE
F1 Test (STAY)	62.5%	45.5%	TRUE
Precision (STAY)	49.4%	44.3%	FALSE
Recall (STAY)	89.2%	58.1%	TRUE
F1 Train (SKIP)	20.6%	45.5%	TRUE
F1 Test (SKIP)	10.7%	35.5%	TRUE
Precision (SKIP)	23.5%	52.4%	TRUE
Recall (SKIP)	9.6%	42.7%	TRUE

Table 4.8: Step 5 (Inter): Proposed Feature (AF7 high gamma mean) vs EI. Significance tested using Wilcoxon signed-rank test against EI (two-sided, $\alpha = 0.05$, $N=25$).

The performance of AF8 delta peak frequency relative to the Engagement Index is shown in Table 4.9. Here, too, the **SKIP** class is predicted more effectively than by the Engagement Index. Precision for **STAY** increases from 49.4% to 52.8%, while for **SKIP**, F_1 -train rises from 20.6% to 54.1%, F_1 -test from 10.7% to 45.5%, Precision from 23.5% to 53.1%, and Recall from 9.6% to 48.7%. Among the three candidate features examined here, AF8 delta peak frequency therefore provides one of the clearest improvements over the Engagement Index in the cross-participant setting. This is particularly important because it suggests that a feature derived from low-frequency timing structure may be more informative for **SKIP** detection than the conventional engagement metric, even if the absolute performance remains moderate. Taken together, these results support the idea that the Engagement Index can capture a limited engagement-related tendency, but that it is not sufficient as a full account of the **SKIP**/**STAY** distinction. More specifically, it appears to be relatively better at reflecting sustained **STAY**-like behavior than at identifying the more transient and behaviorally decisive **SKIP** state.

This creates a clear next question for Section 4.5: what neurophysiological or behavioral mechanism is actually being captured by the Engagement Index, and which of the more specific features are better aligned with the underlying decision process?

Metric	EI LR	AF8 delta peakfreq	
	Val	Val	Sig
F1 Train (STAY)	62.0%	57.4%	TRUE
F1 Test (STAY)	62.5%	48.3%	TRUE
Precision (STAY)	49.4%	52.8%	TRUE
Recall (STAY)	89.2%	54.6%	TRUE
F1 Train (SKIP)	20.6%	54.1%	TRUE
F1 Test (SKIP)	10.7%	45.5%	TRUE
Precision (SKIP)	23.5%	53.1%	TRUE
Recall (SKIP)	9.6%	48.7%	TRUE

Table 4.9: Step 5 (Inter): Proposed Feature (AF8 delta peakfreq) vs EI. Significance tested using Wilcoxon signed-rank test against EI (two-sided, $\alpha = 0.05$, N=25).

4.5 Discussion

The results above do not yet justify strong causal claims about underlying brain mechanisms, but they do point to several robust interpretive directions. Most importantly, the analyses suggest that the conventional Engagement Index is not wrong in a trivial sense; rather, it appears to capture a limited and asymmetrical subset of the behaviorally relevant signal. In particular, it seems more aligned with sustained **STAY**-like states than with the more transient, decision-like **SKIP** events. This asymmetry is crucial, because it implies that the Engagement Index may reflect a broad state of ongoing

engagement, but not the full cognitive transition that leads to disengagement. From a scientific perspective, this is a more useful and humble conclusion than simply saying that the index "works" or "does not work": it works partly, in a constrained way, and its failure modes are themselves informative.

A second important point is that the strongest features are not distributed uniformly across the EEG. Instead, they cluster in specific electrodes, frequency bands, and feature types, with raw broadband variability at TP10, high- γ activity at AF7, and delta peak frequency at AF8 repeatedly emerging as informative. This pattern suggests that the relevant signal may not be a single abstract "engagement" quantity, but rather a combination of lateralized, frequency-specific, and task-sensitive components. The fact that raw broadband signal sometimes outperforms band-limited features also raises the possibility that the discriminative information is not fully captured by conventional band definitions alone. At the same time, this may also indicate sensitivity to non-neural factors, including motion-related or muscle-related artifacts, which must be treated cautiously rather than ignored.

4.5.1 Neuroscience

From a neuroscientific perspective, the most consistent result is that separability increases when the signal is described more specifically, especially in the raw broadband TP10 feature and in AF7 high- γ activity. This may indicate that the decision to continue scrolling versus disengage is associated with a distributed and frequency-sensitive pattern rather than with a single canonical engagement rhythm. The repeated prominence of high- γ and delta-derived features is notable because these ranges are often linked to sharply different physiological processes; however, the present data do not allow a definitive mapping from feature to neural mechanism. What can be said more cautiously is that the observed pattern is compatible with the idea that the task engages both rapid sensorimotor dynamics and higher-order control or monitoring processes, with anterior frontal and temporal sites contributing differently to the observed class structure.

At the same time, the broadband superiority of TP10 and the strong performance of AF7 in high-frequency ranges create an important interpretive friction point. If the signal were purely reflecting a clean cortical engagement index, one might expect a more stable advantage for the conventional composite metric. Instead, the data imply that narrower and more local signal components contain more class-relevant information. This opens at least two competing hypotheses: either the task truly recruits region-specific neurophysiological signatures that the Engagement Index averages away, or the apparent gain is partly driven by peripheral contamination, such as facial

muscle activity, motion, or device-dependent noise.

This pattern is consistent with established evidence that temporal electrodes are disproportionately susceptible to EMG contamination from the underlying temporalis muscle, which persists even at rest and is not fully removed by standard ICA-based artifact correction [23, 25, 36]; because reported EMG beta-peak power extends as low as 13 Hz at frontal and temporal sites [23], this ambiguity is not confined to the high-gamma and broadband features discussed above, and plausibly extends to alpha- and beta-band features at these same electrodes. Because two independent Muse S device units were used across participants (Section 3.2), device-dependent differences in electrode contact or impedance could further modulate this susceptibility, compounding rather than resolving the ambiguity between neural and myogenic origin [24].

A related and separate source of ambiguity concerns the decision to report all results on un-notched data (Section 3.6). Because the 50 Hz mains frequency falls inside the high-gamma band, and because low-cost dry-electrode amplifiers are additionally susceptible to mains-driven intermodulation artifacts flanking the line frequency [29], any high-gamma or broadband finding reported here (e.g. AF7 high-gamma mean, TP10 raw standard deviation) cannot be fully dissociated from possible line-noise contamination. This is an acknowledged limitation of the present analysis rather than a validated non-issue: a targeted notch-versus-no-notch ablation on the reported top features is left to future work.

The delta result is consistent with a broader literature linking delta oscillatory responses to decision-making and attentional processing in awake cognitive paradigms, with reduced delta responses treated as a candidate marker of impaired cognitive processing [26]. From an evolutionary-functional perspective, delta oscillations have also been proposed to index salience detection and basic attentional/motivational processing, rising in relative prominence precisely when higher-frequency, more operationally flexible systems are less engaged or less effective [27] — a pattern plausibly relevant here given the Engagement Index’s weak SKIP-detection performance despite relying primarily on beta, alpha, and theta power. This literature has not previously been applied to short-form video engagement classification specifically; the only directly comparable prior evidence comes from studies with substantially different scope: an adolescent (mean age 12.3) SVA sample with a small, borderline-significant delta–executive-function link ($F=2.184$, $p=.040$) [28], and a smaller ($n=10$), coarser binary phone-use classifier that found beta and gamma, rather than delta, as strongest predictors using different gamma-band boundaries [4]. Neither study addresses a self-paced skip decision, underscoring that the present question is novel rather than merely

under-replicated. The present AF8 delta peak-frequency finding is best read as consistent with delta’s established general role in attention and salience, rather than as a validated TikTok-specific marker.

The electrode pattern also raises a lateralization question that cannot yet be answered decisively. AF7 and TP10 appear more informative than AF8 and TP9, which may point to a functional asymmetry in how engagement-related and action-related states are expressed during naturalistic scrolling. However, because two physical units of the headband were used and because broadband effects may partially reflect artifact structure, this should be interpreted as a hypothesis for future work rather than as a confirmed hemispheric specialization. In the next step, a targeted analysis of artifact-sensitive channels, spectral decomposition, and device-specific effects would be needed to determine whether these patterns reflect genuine neurophysiology or measurement confounds.

4.5.2 Psychology

Psychologically, the results suggest that the act of skipping a TikTok video is not simply the inverse of being engaged. The Engagement Index performs better for **STAY** than for **SKIP**, which implies that sustained attention or passive continuation may be easier to detect than the brief decision point at which a participant chooses to move on. This distinction matters because it suggests that **SKIP** is not merely a low-engagement state, but may represent a qualitatively different behavioral event: an active choice, a threshold crossing, or a moment of disengagement that is preceded by a narrower and more transient signal profile. The stronger performance of features such as AF8 delta peak frequency and TP10 raw standard deviation supports the idea that the decisive moment may be better captured by fine-grained temporal structure than by a broad engagement aggregate.

The burst analysis adds an additional psychological layer. Some participants behave as “fast scrollers” with repeated burst blocks, whereas others show mostly isolated swipes. This heterogeneity implies that the psychological process underlying scrolling is not uniform across participants. Some users may evaluate content quickly and repeatedly, while others may linger, reflect, or maintain a more stable engagement rhythm. That individual variability likely explains why per-participant and cross-participant performance differ so strongly, and why the same feature may appear informative in one setting but unstable in another.

The modest but consistent success of individual features over the Engagement Index also suggests that a psychologically meaningful model of scrolling may need to distinguish between sustained engagement, momentary decision

pressure, and motor execution. The present results are compatible with such a layered interpretation, but they do not yet prove it. What they do show is that an aggregate engagement score is too coarse to explain the full behavioral variance, especially when the behavior is embedded in a rapid, naturalistic, and self-paced interaction environment.

4.5.3 Behaviour

From a behavioral standpoint, the strongest conclusion is that the **SKIP/STAY** decision is heterogeneous and temporally structured. The burst analysis shows that many swipes occur in clusters rather than as isolated events, which means that the observed behavior is not simply a sequence of independent binary decisions. Instead, it appears to reflect a mixture of cautious sampling, rapid dismissal, and occasional repeated disengagement. This helps explain why feature-level models can outperform the Engagement Index in some cases: the neural signal may be tracking not only whether a participant is engaged, but also where they are in a short behavioral sequence leading toward or away from a swipe.

The fact that raw TP10 variability and AF8 delta peak frequency improve **SKIP** detection suggests that the behavioral decision to move on may leave a measurable trace in the EEG, but only weakly and only under certain feature definitions. The most defensible interpretation is therefore not that the model “predicts choice” in a strong sense, but that it detects behavioral precursors or correlates of the **SKIP** decision with limited but meaningful sensitivity. That is an important distinction for the thesis: the signal is informative, but it is not deterministic.

The results also show that behavior is strongly shaped by parameter choices, especially in the burst analysis. Changing the window definition changes how events are grouped, and changing the feature definition changes which patterns emerge as separable. This dependency implies that future behavioral analyses should not rely on a single segmentation or metric, but rather test whether the observed structure is stable across multiple reasonable parameterizations.

4.5.4 Application

In applied terms, these findings are best understood as a proof of concept for more transparent and behaviorally grounded engagement modeling. Rather than demonstrating a ready-to-deploy predictive system, the results show that naturalistic scrolling behavior can be measured with sufficient structure to support interpretable feature-level analysis, and that some EEG-derived

features outperform the conventional Engagement Index in ways that are meaningful for both explanation and future application.

The practical relevance of these findings lies less in the idea of a fully deployable EEG-based prediction system and more in the methodological and conceptual direction they provide. The present results show that naturalistic scrolling behavior can be measured, segmented, and partially explained using a compact set of EEG-derived features, and that these features can outperform a conventional aggregate index in ways that are interpretable at the level of signal structure. This matters because it suggests a path toward more transparent models of engagement: rather than relying on a single composite score, future systems may be built from a small number of physiologically meaningful components that are easier to interpret, debug, and adapt.

One immediate application is in the optimization of message delivery in health communication and behavior-change interventions. If certain EEG signatures are more associated with continued attention or imminent disengagement, then intervention content could be structured to better match those moments of attentional stability or vulnerability. In principle, this could help improve the timing, length, or sequencing of health messages, public-health prompts, educational micro-content, or persuasive interventions in digital environments. For example, short-form health information could be adapted to identify when a user is likely to continue engaging versus when they are likely to skip, which could inform both content design and delivery timing – though at present this discrimination remains modest and, in the cross-participant setting, does not yet exceed the chance-equivalent F_1 reference established in Section 4.4.2.

A second application is in the development of explainable and lightweight screening tools for user engagement in naturalistic settings. Because the strongest results here emerged from a small number of features rather than a large black-box model, the present work may support the design of interpretable real-time monitors that are more feasible than complex deep-learning approaches. Such monitors could be useful in user-testing environments, educational research, human-computer interaction, or cognitive ergonomics, where the goal is not necessarily to make a diagnostic prediction but to identify when a stimulus fails to hold attention or when a participant shifts into a disengaged state.

At the same time, the present results also define the limits of immediate application. The predictive performance is not yet strong enough to justify high-stakes individualized deployment, and the possibility of artifact-related contributions means that any translational use would require substantial validation, device standardization, and robustness testing. In particular, the strong effects observed at TP10 and AF7 raise the possibility that some

predictive power may arise from non-neural sources such as jaw movement, facial tension, or other task-correlated artifacts – a risk that, per Section 4.5, plausibly extends into alpha- and beta-band features at these electrodes and not only into the high-gamma and broadband features discussed above. For this reason, the present study is best interpreted as a proof of principle for explainable signal extraction rather than as a finished applied system.

Recommendations for Policy Makers

Three concrete regulatory implications follow from the present findings without requiring any claim beyond what the data and the cited literature support.

First, **watch time is the wrong regulatory quantity**. Both platform self-reporting and emerging digital-services regulation frequently treat viewing duration as a proxy for user interest or content value. The present results, together with the behavioral literature, indicate that continuous non-skip viewing is equally consistent with dissociative, non-encoding consumption as with genuine engagement [2, 3]; a metric that cannot distinguish the two should not be used as evidence of user benefit in transparency reports or audit frameworks. Policy makers evaluating platform disclosures should therefore require engagement measures that are validated against independent behavioral or physiological criteria rather than dwell time alone. This recommendation follows from the cited dissociation and flow literature [2, 3] rather than from the present EEG results directly, which characterize a candidate physiological signature of the pre-swipe state but do not themselves measure dissociation, flow, or recall.

Second, **friction is an evidence-backed lever**. Interface features that segment or interrupt continuous scrolling — custom lists and reading-history labels, in particular — have been shown to significantly reduce self-reported dissociation without removing content [3] (a single study, on a Twitter client rather than short-form video; replication on TikTok-like platforms is needed before generalizing). Mandating or incentivizing such segmentation cues in feed-based products is a realistic, minimally invasive intervention whose mechanism is already empirically characterized, in contrast to blunt instruments such as age-gating or time caps.

Third, **auditability of engagement optimization**. The present thesis demonstrates that the neural precursors of the skip decision are partially measurable with consumer-grade hardware. Platforms plausibly have access to behavioral, and possibly physiological, signals at a scale and precision this thesis cannot match; if so, the same auditability logic applies with greater force. This thesis has no access to platform-internal data, and this claim

should be read as a reasonable extrapolation rather than a demonstrated fact. Regulators should treat engagement-optimization systems as auditable algorithmic systems and require disclosure of the behavioral targets they optimize, precisely because the optimized quantity (continued viewing) is, per the above, not a wellbeing proxy.

Recommendations for Digital Global Public Health Research

For digital global public health — including the research agenda of the HPI Chair for Digital Global Public Health, whose stated priorities include disease prevention, evidence-based health communication, and precision public health — the present work suggests three tangible extensions.

First, **message pre-testing panels**: the full recording and analysis pipeline of this thesis runs on a consumer headset, a laptop, and open-source software, and is reproducible by an independent researcher (Section 4.2). This makes small- n physiological pre-testing of health messages — comparing candidate formats of, e.g., a vaccination or antimicrobial-resistance communication for their pre-skip signature before campaign deployment — more accessible to public institutions than fully custom or clinical-grade EEG pipelines would be, analogous to the message pre-testing already standard in commercial advertising but at research-grade transparency — though, as Section 4.2 notes, substantial engineering investment was still required to reach this point, and this should be read as a reduction in barrier-to-entry rather than a turnkey solution.

Second, **attentional reach as a campaign endpoint**: public health communication on short-form platforms is known to underperform entertainment content in engagement metrics, with institutional accounts reaching orders-of-magnitude fewer users than individual creators [37]. Campaign evaluation should therefore report not only impressions but survival within the feed — the probability of a health message being skipped within its first seconds — a quantity the present labeling framework provides an initial, currently modest-power operationalization of (Section 4.4.2).

Third, **precision public health applied to attention**: in the same way precision public health stratifies prevention by risk profile, the inter-individual heterogeneity observed here (“frantic” versus “careful” scrollers, Section 4.3; the failure of cross-participant models relative to within-participant models) implies that attentional response to health content is itself a stratifiable trait, and that one-size-fits-all message design leaves systematic reach on the table.

Recommendations for Health Message Design

At the level of individual message creation, the findings translate into concrete design guidance.

(i) **The decision plausibly precedes the swipe:** the SKIP-predictive window examined here (2–5 seconds before the physical action) was adopted as a literature-motivated heuristic starting point rather than an independently validated decision boundary (Section 3.4.1), and class separability within it, while statistically distinguishable from the Engagement Index baseline, remains modest in absolute terms (Section 4.4.2). With that scoping in mind, the general pattern – that some pre-swipe neural signature precedes the physical swipe action – is broadly consistent with content needing to establish interest early rather than relying on a delayed payoff; testing where within a video’s opening seconds this signature is strongest, and whether it maps onto specific content features, is a natural extension of the present labeling framework rather than a claim established by it here.

(ii) **Design for the dissociated viewer:** because a substantial fraction of viewing occurs in a passive, low-encoding state [2, 3], message formats should include pattern interrupts — changes in pacing, framing, or sensory structure — intended to manufacture re-engagement points rather than rely on spontaneous recovery of attention, since Baughan et al. found such recovery is often only recognized retrospectively, if at all [3].

(iii) **Do not optimize health content for dwell time:** a message that is watched to completion in a dissociative state may be encoded worse than a shorter message that triggers active appraisal; evaluation should weight recall and behavior change over watch-through rate.

(iv) **Respect burst structure:** swipes cluster into rapid bursts for a substantial share of events, though the exact proportion is parameter-dependent – 67.4% of swipes are isolated singletons under the default ± 3.0 s window, falling to 48.9% under a wider ± 5.0 s window (Section 4.3) – and burst tendency varies sharply between participants (SD of 16–21 percentage points around these means), consistent with the individual heterogeneity noted in Section 4.5. This favors repeated, distributed placement over single high-cost placements as a population-level default, while flagging burst-aware, per-user delivery timing as a natural extension once outcome-linked validation is available; the present work establishes that swipes cluster, not that burst membership itself predicts weaker message encoding or evaluation.

Societal Scope: An Explicitly Speculative Outlook

The following paragraph is a deliberate change of register: it is a hypothesis-generating extrapolation, not a result of this thesis, and is marked as such. What the data of this thesis and the cited literature jointly establish is narrow but non-trivial: short-form feeds occupy users in a high-frequency stream of appraisal micro-decisions; a substantial share of the resulting viewing time is spent in states of reduced self-awareness and poor encoding [2,3]; and attention is a finite, fluctuating resource whose allocation determines what is subsequently remembered [1]. From these premises, a *time-displacement hypothesis* follows naturally, though it is not tested here: hours spent in dissociative feed consumption are hours of attentional and social bandwidth not spent elsewhere — including on the effortful, in-person interactions through which relationships are formed. Whether this displacement contributes causally to widely discussed, though contested, population-level trends in youth socialization, loneliness, or, further downstream, family formation is precisely the kind of long-range causal chain the present study can neither support nor refute, and this thesis makes no such claim. What it can state is the conditional: *if* such displacement effects exist, then the mechanisms studied here — the moment of attentional lock-in, the dissociative default, the decodable pre-skip state — are the proximate levers through which they operate, and the same measurement framework used here to describe them is the framework with which mitigation (segmentation cues, adaptive interruption, attention-aware content policy) would be evaluated. Quantifying the displacement chain, and testing realistic mitigations against it, is left to future research; the contribution of the present work is to make its first link measurable.

Finally, a dual-use note is warranted: any marker that helps health communicators survive the feed helps commercial and political actors equally. The appropriate response is not to withhold such research — platforms already possess strictly superior behavioral data — but to conduct it transparently, under ethics oversight (Section 3.1), and to pair it with the auditability recommendations above.

4.6 Future Research

Future research should first focus on validation and replication. The present findings are promising, but they are based on a single dataset with a relatively small sample size and two physical units of the same recording device. Replication in an independent sample is therefore essential, ideally with more

uniform hardware, tighter artifact control, a broader and less academically-skewed participant pool, and multiple-comparison-corrected significance testing across the feature–baseline comparisons reported here. A key priority should be to test whether the same features – especially TP10 raw standard deviation, AF7 high- γ mean, and AF8 delta peak frequency – remain informative under repeated acquisition, different content types, and different user populations.

A second major direction is mechanistic clarification. The present results leave open whether the strongest effects reflect genuinely neural engagement processes, movement-related artifacts, or a mixture of both. Future work should therefore include explicit artifact analyses, such as electromyographic control across the alpha-through-gamma range (not only high-gamma or broadband), motion monitoring, ICA-based comparison, channel-quality assessments, and device-specific sensitivity checks. It would also be valuable to compare broadband features against strictly cleaned, source-reconstructed, or artifact-masked data to determine how much of the predictive signal survives after conservative preprocessing.

A third direction concerns theory development. The Engagement Index appears to capture a limited but real component of sustained engagement, yet it may not capture the behavioral threshold that leads to a swipe. Future studies should therefore distinguish more explicitly between sustained attention, momentary decision pressure, action readiness, and actual behavior. This could be pursued by combining EEG with behavioral timing, self-report, eye tracking, or post-hoc content ratings, and should include testing whether burst membership itself (Section 4.3) predicts weaker encoding or evaluation, rather than merely describing swipe clustering. In particular, event-aligned designs could help determine whether the relevant neural signature precedes disengagement, coincides with motor preparation, or reflects post-decisional processing.

A fourth opportunity is methodological expansion. The current pipeline is deliberately simple and interpretable, but future work could test whether other transparent models improve both performance and explanatory value, such as regularized generalized linear models, hierarchical models, or hybrid approaches that preserve interpretability while capturing nonlinearity. At the same time, more elaborate feature engineering may be worthwhile if it remains conceptually grounded; examples include temporal derivatives, burst-aware features, cross-frequency interactions, or individualized feature selection. However, this should be pursued carefully, because greater complexity can easily reduce interpretability and increase the risk of overfitting.

A fifth and highly relevant direction concerns public health and behavior change. Short-form digital media is increasingly used not only for entertain-

ment but also for public communication, education, and health promotion. If EEG-informed engagement models can eventually be made robust and explainable, they may support the design of interventions that adapt message format, timing, and length to the user’s attentional state. This could matter for smoking cessation messages, vaccine communication, mental-health micro-interventions, nutrition prompts, physical-activity nudges, and other digital behavior-change tools. The challenge is not only technical but also ethical: such systems must avoid manipulation, respect autonomy, and be transparent about how engagement is inferred and used.

Finally, future research should examine generalizability across contexts, populations, and content domains. The present sample skewed toward an academically-oriented population recruited via university mailing lists (Section 3.1), and the present task used naturalistic TikTok browsing; similar engagement dynamics may appear in other forms of media consumption, educational content, or digital decision tasks, but this would need to be tested in a broader participant pool before being assumed. Testing whether the same feature families generalize to different content domains would help determine whether the observed effects are specific to short-form video or whether they reflect a broader principle of human engagement and disengagement. If replicated, this line of work could contribute not only to neuroscience and psychology, but also to media research, user-experience design, and public-health communication.

Overall, the present study opens a broad but carefully bounded research program: one that combines explainable EEG analysis, naturalistic behavior, and applied communication research. Its value lies not in claiming a final solution, but in identifying a credible path toward more interpretable, behaviorally grounded, and potentially useful models of engagement and disengagement in real-world digital environments.

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